nature machine intelligence

Article

https://doi.org/10.1038/s42256-026-01293-x

Causal evidence that language models use confidence to drive behaviour

Received: 12 December 2025

Accepted: 20 July 2026

Published online: 07 September 2026

Check for updates

Dharshan Kumaran[1], Nathaniel Daw[1,2], Simon Osindero[1], Petar Veličković[1] & Viorica Patraucean[1]

Metacognition—assessing the quality of one's own cognitive performance—guides adaptive behaviour across species. Confidence signals can be extracted from language model outputs, yet a fundamental question remains: do models actually use these signals to decide whether to answer or abstain? Here we developed a four-phase paradigm. Phase 1 elicited baseline confidence without an abstention option. Phase 2 showed that large language models apply an implicit threshold to internal confidence when abstaining, with confidence effect sizes roughly an order of magnitude larger than alternative mechanisms. Phase 3 provided causal evidence via activation steering: boosting or suppressing confidence correspondingly decreased or increased abstention, with mediation analysis confirming confidence redistribution as the primary mechanism. Phase 4 instructed models to abstain at different confidence levels and showed they adjusted their behaviour accordingly, indicating they read out and act on confidence to set abstention policies. Critically, verbal confidence—an explicit self-evaluation from a separate forward pass—independently predicted abstention across all models despite being less discriminatory of correctness, and activation decoding showed both measures (that is, token log probabilities and verbal confidence) are lossy read-outs of a richer internal representation. Together, these results suggest that abstention is captured by the joint operation of a multidimensional internal confidence representation and threshold-based policies—consistent with structured metacognitive control in large language models—a capacity of growing importance as models transition to autonomous agents that must recognize their own uncertainty.

Humans and animals use confidence—that is, an internal estimate of the probability that a decision is correct—to guide adaptive behaviour[1–3]. Low confidence, for example, can drive a tendency to change one's mind, or gather more information[4,5]. High confidence in a decision, by contrast, can motivate planning and sequential decision-making in the resulting scenarios. Although there is a wide body of research investigating whether large language models (LLMs) can produce confidence ratings that are aligned with ground-truth probabilities[6–8], little work has been carried out to explore whether LLMs themselves can utilize an internal sense of confidence to guide their own decisions—a hallmark of metacognition. Metacognitive control, the ability to use confidence signals to regulate behaviour based on monitoring one's own cognitive states[2,9], has been extensively documented in humans and animals, but remains poorly understood in artificial systems.

Here we focus on one scenario in which confidence plays a key role: deciding whether to answer a question, or instead abstain. This paradigm is inspired by psychophysical and neuroscience studies of decision-making in which evidence suggests that animals use a sense

[1]Google DeepMind, London, UK. [2]Princeton University, Princeton, NJ, USA. e-mail: dkumaran@google.com

 https://doi.org/10.1038/s42256-026-01293-x

of internal confidence to choose whether to take a test or wait for a performance-contingent reward[2,10–12]. We were motivated, therefore, to study LLM abstention as a fundamental expression of a metadecision—that is, a decision about a primary decision—specifically, whether to commit to an answer or withhold it based on internal confidence[2]. In particular, the capacity of LLMs to abstain is also of substantial importance to their safe operation in the real world, since low-confidence—and, therefore, likely incorrect—answers to high-stakes questions (for example, in the medical domain) are usually more harmful than refusals. Although emerging research has demonstrated the use of confidence for the abstention and refusal of harmful content, these typically rely on post hoc thresholding, or specialized fine-tuning procedures[13–22]. Clear evidence that LLMs can instead exploit their native internal sense of confidence to guide abstention decisions remains lacking. More specifically, it remains unclear whether such behaviour can be entirely explained in terms of confidence signals tied directly to answer generation (that is, token log probabilities), or whether a more evaluative confidence signal (verbal confidence) also contributes[9].

We developed a controlled paradigm to investigate whether and how confidence signals drive abstention behaviour. In Phase 1, LLMs completed four-option multiple-choice questions drawn from a factuality dataset (Fig. 1). We measured two forms of confidence during Phase 1: calibrated log-probability-based confidence derived from the model's output distribution, and verbal confidence—an explicit self-report of certainty elicited in a separate pass. Both served as pre-decisional measures, uncontaminated by the presence of an abstention option, that we used to model and predict abstention behaviour across subsequent phases. In Phase 2, the same questions were presented, but with abstention now available as a response. In Phase 3, we tested causality by experimentally modulating the abstention rates through the enhancement or suppression of internal confidence signals. Finally, in Phase 4, models were instructed to abstain whenever their confidence fell below a specified threshold, which we systematically varied. Additionally, to characterize the internal basis of these confidence signals, we used activation decoding at the pre-answer token—the locus used for activation steering—to test whether both measures are partial read-outs of a richer internal confidence representation.

Our paradigm, therefore, allowed us to obtain causal evidence for the role of confidence signals at two key stages in the decision-making process (Fig. 1): (1) the confidence representations themselves, and (2) the policy governing how confidence is used to guide abstention decisions. Our two-stage framework draws on research demonstrating that human and animal metacognitive decisions involve separable stages: confidence formation and threshold-based action selection[1,2,9,12,23,24]. Although classic two-stage models focus on how confidence emerges from evidence accumulation, here we take confidence representations as already formed and focus on how they are deployed to guide metadecisions. In particular, this two-stage framework operates at a computational level of analysis, characterizing the functional architecture of confidence-guided metadecisions: specifically, what problem is being solved (mapping confidence to abstention) and the key components involved—without making claims about the specific neural or algorithmic implementation in transformer architectures.

Results

Phase 1: basic performance and calibrated confidence (GPT-4o)

We focus on GPT-4o for the main analyses, with the exception of activation steering (Phase 3), which requires access to internal activations and was, therefore, conducted using Gemma 3 27B. Detailed results for Gemma 3 27B, DeepSeek 671B and Qwen 80B are reported in Supplementary Results.

GPT-4o performed at 63.7% correct during Phase 1 (four-way multiple-choice, no abstention option; Methods). Following standard practice[25], we calibrated the model's logits on a separate dataset using temperature scaling (optimal temperature = 4.1; expected calibration error (ECE) = 0.046; area under the receiver operating characteristic curve (AUROC) = 0.90; Methods). Throughout, we refer to these temperature-scaled probabilities as calibrated confidence. This measure serves as an external read-out of the model's internal confidence state; in later sections, we additionally examine verbal confidence—an explicit self-report of certainty—as a second, partially independent read-out.

As expected, calibrated confidence strongly predicted error rate on a separate Phase 1 test dataset ($r(12) = -0.97$, $P = 7.96 \times 10^{-9}$; Fig. 2a,b), confirming that the model's logits provide a meaningful signal of choice accuracy.

Phase 2: calibrated confidence predicts abstention (GPT-4o)

In Phase 2, the same questions included an abstention option (Methods and Fig. 1), yielding 30.0% correct, 13.4% incorrect and 56.6% abstention. Accuracy among answered questions increased from 63.7% to 69.1%, consistent with a coverage–accuracy trade-off (Supplementary Fig. 1).

We used Phase-1-calibrated confidence to model Phase 2 abstention behaviour within a two-stage confidence–decision framework (Fig. 1): Stage 1 represents the formation of confidence representations and Stage 2, a threshold-based policy mapping confidence to abstention decisions. Phase 1 confidence provides a pre-decisional measure uncontaminated by the presence of an abstention option, allowing a clean test of whether confidence representations formed independently generalize to drive abstention when that option becomes available. At the same time, it remains strongly informative about later behaviour: the Phase 1 chosen option remained the highest-confidence real alternative in 82.0% of Phase 2 trials and 81.6% of Phase 4 trials. In particular, the cross-phase stability of the real-option confidence distribution was higher on answered trials than on abstention trials in Phase 2 (Supplementary Results and Supplementary Table 2), consistent with a two-stage process in which the model first evaluates the real options and then reassesses when internal confidence falls below an implicit threshold—a mechanism we validate directly in Phase 4 using externally specified thresholds.

However, several alternative attributes could potentially guide abstention. The model might rely on question difficulty—an objective measure of item hardness independent of subjective confidence on a particular trial (that is, mean correctness across multiple samples; see Methods). Abstention could also be driven by knowledge retrieval accessibility, measured through retrieval-augmented generation (RAG) scores[26], or surface-level linguistic features captured through sentence embedding similarity scores[27,28].

Phase 2: computational modelling of abstention behaviour (GPT-4o). We used logistic regression to model binary abstention choices, quantifying the relative contribution of calibrated confidence, question difficulty, knowledge retrieval accessibility (RAG scores) and surface-level semantic features (sentence embeddings) (Methods and Supplementary Table 1).

Calibrated confidence alone was a strong predictor of abstention (Akaike information criterion (AIC) = 1,227.4, pseudo-R^2 = 0.106, $\beta_C = -4.87 \pm 0.47$, $z = -10.37$, $P = 1.3 \times 10^{-29}$), substantially outperforming question difficulty (AIC = 1,346.6, pseudo-R^2 = 0.019), RAG scores (AIC = 1,365.6, pseudo-R^2 = 0.005) and sentence embeddings (AIC = 1,368.2, pseudo-R^2 = 0.017) (Fig. 2c, left). In the combined confidence + difficulty model, difficulty became marginal ($P = 0.053$), confirming that abstention is primarily driven by calibrated confidence rather than objective item hardness. In practical terms, a 0.1-unit increase in confidence reduces the probability of abstaining by approximately 12 percentage points. Likelihood ratio tests confirmed that confidence added substantial explanatory power above each alternative predictor (ΔAIC > 119 in all cases; Supplementary Results shows the exact two-sided P values), whereas the reverse was not true: neither RAG nor difficulty added once confidence was included (Supplementary Results for details of the full likelihood ratio test).

In the full model including all predictors simultaneously, calibrated confidence remained overwhelmingly dominant

https://doi.org/10.1038/s42256-026-01293-x

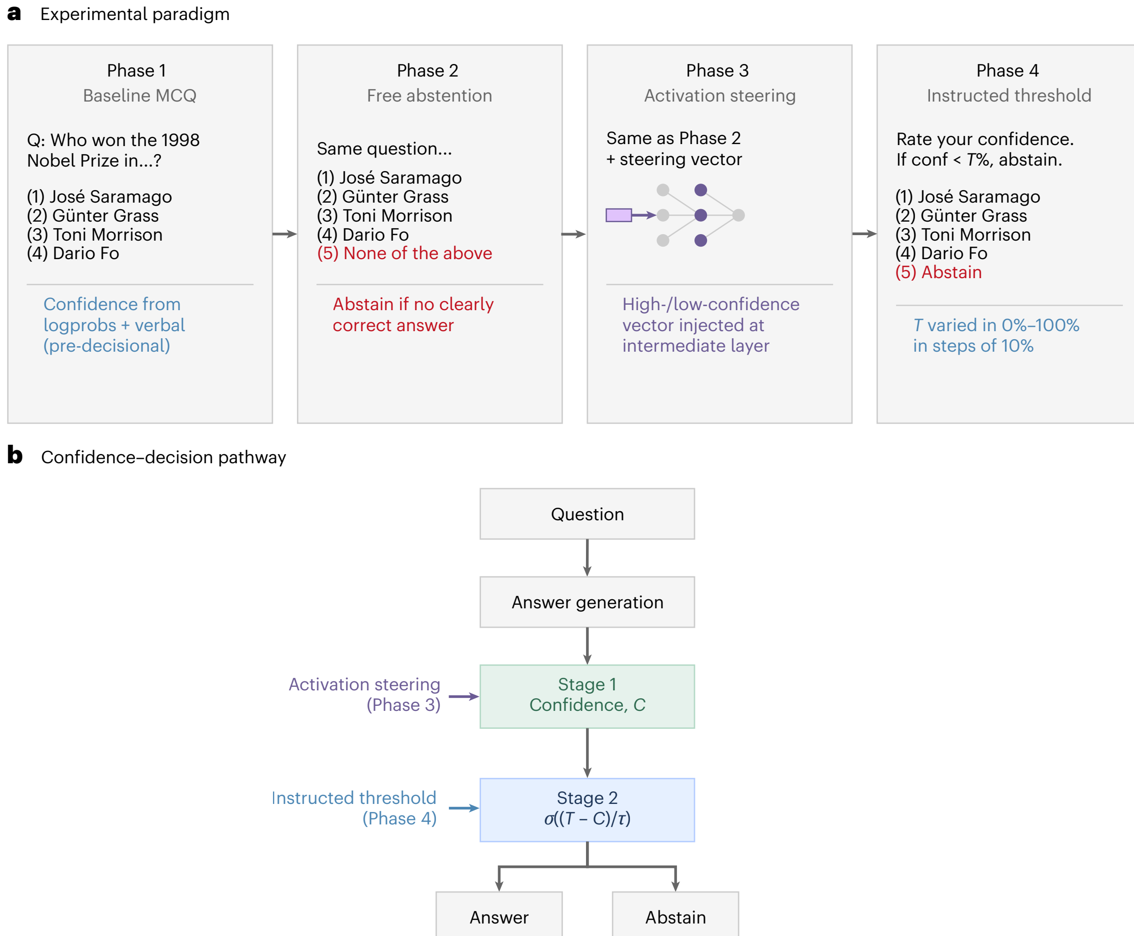

Fig. 1 | Overview of the four phases of the abstention paradigm, and confidence–decision pathway. a, In Phase 1, the model had to choose between four real options, with no abstention option. The inclusion of Phase 1 allowed us to obtain the log-probability-based and verbal confidence ratings from the model uncontaminated by the presence of an abstention option/decision. Phase 1 chosen confidence was used to model behaviour in Phases 2 and 4. In Phases 2 and 3, the same questions included an abstention option, with models instructed to abstain when no clearly correct answer was apparent—with activation steering applied in the latter. In Phase 4, the model was given specific instructions to first choose an answer and then rate its confidence in its answer—and finally output its answer if its confidence was greater than a threshold, T. If its confidence was lower than T, it was instructed to abstain. Note that these are illustrative prompts; the full prompts used are provided in Methods. MCQ, multiple-choice questions. **b**, In Stage 1, the model generates an answer and associated confidence signals C. In our abstention paradigm, confidence signals drive behaviour by being compared with a threshold T with the decision process further parameterized by a policy temperature τ (Stage 2). Although we separate answer generation from confidence signals in our schematic, confidence signals may be generated as a direct consequence of answer generation—as is the case in first-order accounts of confidence[9,50].

(Fig. 2c, right). The standardized effect of confidence ($|\beta_{std}| = 0.99$) was approximately an order of magnitude larger than RAG, difficulty or the strongest embedding component, confirming that abstention behaviour is driven by calibrated confidence rather than by alternative mechanisms.

From the logistic model, we recovered the implicit decision parameters underlying GPT-4o's abstention policy (Methods). The indifference point T_{50}—the confidence level at which the model abstains 50% of the time—was approximately 77%, indicating that GPT-4o requires substantial confidence before choosing to answer. The policy temperature (20 confidence units) indicates a soft, probabilistic transition rather than an all-or-nothing rule (Fig. 2d), paralleling patterns of confidence-based decision-making in humans[2]. Although our focus here is GPT-4o and calibrated confidence, qualitatively similar findings were observed in all other models tested and using verbal confidence (see below and Supplementary Results).

Phase 3: activation steering (Gemma 3 27B)

Phase 2 established that calibrated confidence predicts abstention. We next sought direct causal evidence by intervening at the level of confidence representations (Fig. 1, Stage 1). Using activation steering[29,30] in Gemma 3 27B, we constructed steering vectors from Phase 2 activations by contrasting trials with high- versus low-calibrated-confidence margins—the difference between the model's confidence in its best answer option and its confidence in abstaining (Methods). At the test time, we injected high- or low-confidence vectors at specific layers during inference on 500 held-out questions. If confidence causally drives abstention, high-confidence steering should reduce abstention rates, and low-confidence steering should increase them.

As hypothesized, activation steering markedly influenced the abstention behaviour (Fig. 3a). The effect peaked at intermediate layers (layer 31; Supplementary Fig. 2). Abstention declined from 66.5% at maximum low-confidence steering to 7.0% at maximum high-confidence steering—a 59.5 percentage point swing (two-sided Pearson correlation, $r(7) = -0.9855$, $P = 1.19 \times 10^{-6}$; averaged across layers 30–40; Fig. 3b).

Steering induced coordinated shifts in the underlying calibrated confidence distribution: high-confidence steering boosted confidence in real answer options and suppressed confidence in the abstention option, and vice versa (Fig. 3c,d). The change in the maximum real

https://doi.org/10.1038/s42256-026-01293-x

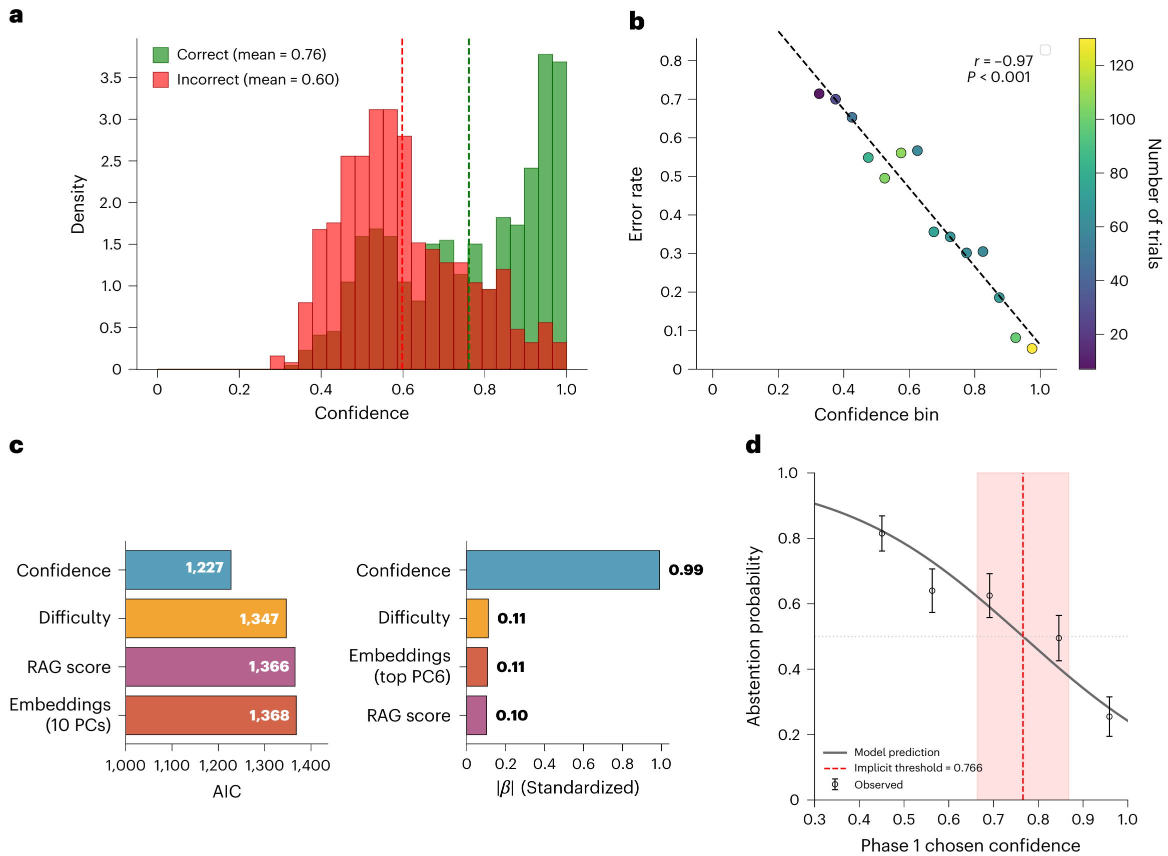

Fig. 2 | GPT-4o-calibrated confidence predicts error rate (Phase 1) and abstention behaviour (Phase 2). a, Calibrated confidence distributions for correct (green) and incorrect (red) Phase 1 responses. Dashed vertical lines denote the mean confidence for each group. **b**, Relationship between binned confidence and error rate in Phase 1. Each point represents 1 confidence bin (width = 0.05; $n = 14$ bins, corresponding to 1,000 independent model queries total on unique SimpleQA questions). Colour indicates the number of trials per bin. Confidence and error rate were strongly negatively correlated (two-sided Pearson correlation, $r(12) = -0.97$, $P = 7.96 \times 10^{-9}$). The model was calibrated on a separate set of 1,000 trials (Methods). **c**, Model comparison showing calibrated confidence as the dominant predictor of abstention. Left: AIC values for single-predictor models, where lower values indicate a better fit. Right: standardized effect sizes from the full model including all predictors. Confidence exhibits an effect size ($|\beta_{std}| = 0.99$) approximately 10 times larger than alternative predictors, demonstrating that abstention is primarily predicted by confidence rather than question difficulty, knowledge accessibility or surface-level semantic features. **d**, Logistic regression model of natural abstention probability as a function of calibrated confidence for $n = 1{,}000$ trials. The grey curve indicates the model prediction from the confidence + difficulty model, holding difficulty at its mean value. The red dashed line marks the implicit decision threshold at 77%, where P(abstain) = 0.5. The red-shaded region denotes the CI spanning approximately ±10 percentage points around the threshold, illustrating the gradual transition of the decision boundary. Black circles represent the observed abstention rates (mean; error bars show 95% Wilson CIs) across confidence quintiles. Logistic regression was fitted to $n = 1{,}000$ independent trials (1 model query per unique SimpleQA question; independent observations). Together, these panels reveal GPT-4o's intrinsic abstention policy: higher calibrated confidence corresponds to a lower likelihood of abstaining, even without explicit threshold instructions.

confidence was strongly anticorrelated with the change in abstention confidence (two-sided Pearson correlation, $r(43, 998) = -0.8943$, $P < 10^{-300}$), indicating that steering operates through a coherent redistribution of confidence from abstention towards answer options. Steering also produced a small coverage–accuracy trade-off: accuracy among answered questions declined from 59.2% to 53.7% as steering shifted from maximum low to high confidence, whereas coverage increased from 33.5% to 93.0% (Fig. 3b).

Mediation analysis. We next asked whether steering influences abstention through changes in calibrated confidence (Stage 1) or through changes in the decision policy mapping confidence to choices (Stage 2). We conducted a parallel mediation analysis (layers 30–40) with two mediators (Methods and Fig. 4): (1) a confidence redistribution mediator—the net shift in calibrated confidence from the abstention option towards real answer options ($\Delta_{\text{max real}} - \Delta_{\text{abstention}}$); and (2) a policy shift mediator—the change in predicted abstention probability between steered and baseline calibration curves, holding confidence constant. These mediators correspond to Stages 1 and 2 of the confidence–decision pathway, respectively. A third, direct pathway captures mechanisms outside both measured mediators.

Confidence redistribution was the dominant mechanism (67.1% of the total effect; Fig. 4). Steering reduced abstention (total effect: $c = -0.82$, 95% CI [−0.89, −0.67]). High-confidence steering increased the net confidence shift towards real options, which, in turn, strongly reduced abstention, yielding an indirect effect of $a_1 \times b_1 = -0.55$ (95% CI [−0.65, −0.47]). A secondary pathway operated through policy changes (26.2% of the total effect; indirect effect $a_2 \times b_2 = -0.22$, 95% CI [−0.24, −0.19]). A residual direct effect remained ($c' = -0.32$), and together, the two indirect pathways accounted for 93.3% of the total effect. Due to the nonlinear logit link function, direct and indirect effects do not sum exactly to the total effect, though relative contributions remain interpretable[31].

These results demonstrate that activation steering operates primarily through Stage 1 confidence redistribution—reallocating confidence from abstention towards answer options—leaving the Stage 2 decision policy largely unchanged. The mediation pattern was robust to the inclusion of item difficulty as a covariate, with

https://doi.org/10.1038/s42256-026-01293-x

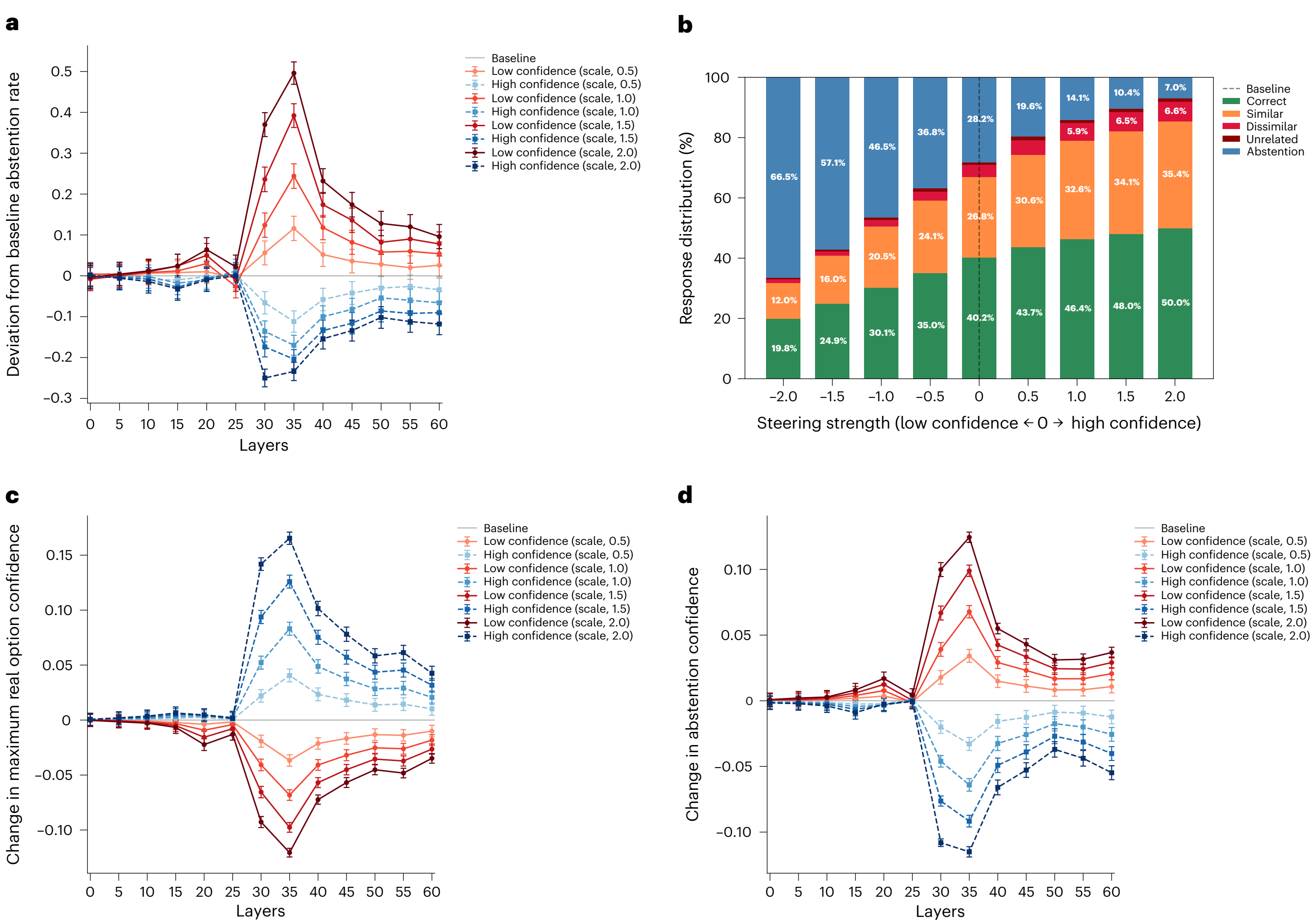

Fig. 3 | Effects of activation steering on abstention behaviour and calibrated confidence in Gemma 3 27B. a, Change from baseline abstention rate as a function of high- and low-confidence activation steering. Values are means across $n = 500$ independent model queries on unique held-out SimpleQA questions; error bars show ±s.e.m. Baseline level of abstention in the no-steering condition is 28.2%. **b**, Response distribution averaged across layers 30–40 as a function of steering strength (from low confidence (negative) to baseline (0) and high confidence (positive)). **c**, Change from baseline-calibrated confidence in the highest-confidence real answer option (regardless of whether a real option or abstention was selected) as a function of steering strength. Values are means across $n = 500$ independent model queries on unique held-out SimpleQA questions; error bars show ±s.e.m. **d**, Change from baseline-calibrated confidence assigned to the abstention option (regardless of outcome) as a function of steering strength. Values are means across $n = 500$ independent model queries on unique held-out SimpleQA questions; error bars show ±s.e.m. Baseline maximum real confidence was 0.37, and abstention confidence was 0.25. Activation steering was evaluated on a held-out set of 500 questions not used to construct the steering vectors.

confidence redistribution accounting for 71.0% of the total effect after controlling for difficulty (Supplementary Results shows the full difficulty-controlled analyses and replication with Gemma-specific difficulty scores). Together, these findings provide direct causal evidence that confidence signals play a primary role in driving abstention behaviour in LLMs.

Phase 4: instructed thresholds modulate abstention behaviour (GPT-4o)

In Phase 4, LLMs were instructed to abstain when their confidence fell below a specified threshold, which we systematically varied. This manipulation directly targets Stage 2 of the confidence–decision pathway, allowing us to test whether altering the policy causally influences abstention rates and whether threshold instructions operate purely at the decision boundary (Stage 2) or also influence internal confidence representations (Stage 1). As expected, increasing thresholds increased the abstention rates (Fig. 5a), with a corresponding coverage–accuracy trade-off: higher thresholds led to more accurate responses among answered trials ($\beta = 0.0051$, $z = 3.97$, $P = 7.27 \times 10^{-5}$; Supplementary Fig. 3 and Supplementary Results).

As in Phase 2, we used Phase-1-calibrated confidence as the predictor of Phase 4 behaviour, since it represents a pre-decisional measure uncontaminated by the threshold instruction. This choice is empirically validated by the structure of abstention behaviour: when plotted against the Phase 1 confidence, abstention exhibits a smooth diagonal gradient, consistent with threshold and confidence acting as independent inputs (Fig. 5b). By contrast, when plotted against Phase 4 confidence, abstention collapses into horizontal bands, indicating that Phase 4 confidence has already incorporated the threshold instruction and reflects post-decisional processing (Fig. 5c; Supplementary Results shows a quantitative analysis). The critical test is whether Phase 1 confidence retains its predictive power across Phase 4—if so, this indicates that instructed thresholds operate at Stage 2 (decision policy), leaving Stage 1 confidence representations fundamentally unchanged.

Phase 4: computational modelling of abstention behaviour (GPT-4o). We fit logistic regression models predicting binary abstention from calibrated confidence, instructed threshold and the same alternative predictors tested in Phase 2 (Fig. 5d, Methods, and Supplementary Tables 3 and 4). Adding confidence to a threshold-only

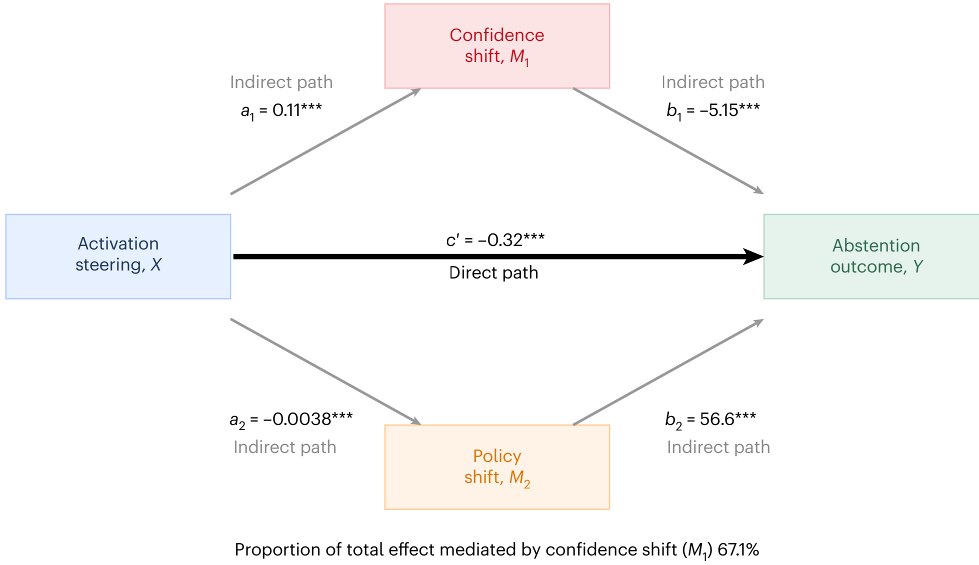

Fig. 4 | Results of Gemma 3 27B parallel mediation analysis: confidence redistribution is the dominant mechanism underlying steering effects on abstention. Path diagram showing how activation steering influences abstention rates through two parallel indirect pathways (confidence redistribution and policy changes) and a direct pathway. Analyses included $n = 44{,}000$ observations from 500 independent held-out questions across 11 layers and 8 steering conditions. The confidence shift mediator (M_1) represents the change in maximum real calibrated confidence minus the change in calibrated abstention confidence induced by steering. The policy shift mediator (M_2) captures changes in the calibration curve mapping confidence to abstention probability. The mediators were weakly correlated (two-sided Pearson correlation, $r(43{,}998) = -0.2383$, $P = 9.58 \times 10^{-300}$). Activation steering significantly increased the net confidence shift towards real options ($a_1 = 0.1073$, $P < 10^{-300}$), which, in turn, reduced the abstention odds ($b_1 = -5.1521$, $P = 2.856 \times 10^{-199}$), yielding an indirect effect of $a_1 \times b_1 = -0.5529$ (95% CI [−0.6426, −0.4733]). Steering had a small but significant effect on the decision policy ($a_2 = -0.0038$, $P = 1.272 \times 10^{-34}$), and policy shifts exerted a strong effect on abstention when they occurred ($b_2 = 56.6080$, $P < 10^{-300}$), yielding an indirect effect of $a_2 \times b_2 = -0.2156$ (95% CI [−0.2427, −0.1905]). A direct effect persisted after controlling for both mediators ($c' = -0.3203$, $P = 8.334 \times 10^{-65}$), relative to a total effect of $c = -0.8240$ ($P < 10^{-300}$). Path coefficients a_1 and a_2 were estimated using ordinary least-squares regression with cluster-robust standard errors clustered by the question; b_1, b_2, c' and c were estimated using binomial generalized linear models. Indirect-effect CIs were estimated using a cluster bootstrap with 1,000 resamples. All reported P values are two sided. Due to the nonlinear logit link function, the direct and indirect effects do not sum exactly to the total effect, although their relative contributions remain interpretable. Confidence redistribution accounted for 67.1% of the total effect, whereas policy shifts accounted for 26.2%. ***$P < 0.001$ indicates that effects are significant.

model more than doubled the explained variance (pseudo-R^2: 0.11→0.24; ΔAIC = −1,953, likelihood ratio test $P < 2.2 \times 10^{-16}$), whereas alternative predictors added negligibly. In the full model, confidence ($|\beta_{std}| = 1.09$) and threshold ($|\beta_{std}| = 1.03$) exhibited comparable standardized effect sizes, both substantially larger than RAG (0.22), difficulty (0.05) or embeddings (0.11) (Fig. 5d; Supplementary Results provides the full model comparison details). These results confirm that Phase 1 confidence retains strong predictive power across all threshold conditions, supporting the operational independence of Stage 1 (confidence formation) and Stage 2 (threshold policy).

From the fitted coefficients, we derived three interpretable decision parameters (Methods). The scale parameter (1.80) reveals that GPT-4o weights its own confidence nearly twice as heavily as instructed thresholds—a 1% increase in confidence offsets approximately 1.80% of threshold. The shift parameter (−97.6) indicates strong baseline conservatism, with the model tending to abstain even at confidence levels above the threshold, consistent with treating errors as costlier than unnecessary abstentions. The policy temperature (31.0) indicates a soft, gradual transition rather than a sharp decision boundary. At 80% confidence, the indifference (instructed) threshold T_{50} is approximately 46% (Fig. 5e). Despite these asymmetric tendencies, the model accurately captured observed abstention rates across all conditions (Fig. 5f).

Phase 4, thus, demonstrates that abstention is well captured by a two-stage model in which pre-decisional confidence (Stage 1) is compared against the instructed thresholds (Stage 2). The strong predictive power of Phase 1 confidence across all the threshold conditions indicates that thresholds operate primarily at Stage 2 without fundamentally distorting Stage 1 representations, although the systematic overweighting of confidence (scale = 1.80) suggests that threshold instructions may influence how confidence is weighted in decisions. Analysis of the continuous Phase 4 abstention confidence provided convergent evidence: threshold shifted the model towards abstention, whereas confidence pulled it back towards answering (Supplementary Fig. 4 and Supplementary Results).

We also tested Gemma 3 27B, DeepSeek 671B and Qwen 80B on Phase 4. The prompt used with GPT-4o elicited few abstentions in Gemma 3 27B, consistent with model-specific differences in instruction adherence[13,32,33]. We, therefore, used a rephrased prompt to ensure meaningful engagement with the abstention paradigm (Methods provides details of the prompt generation procedure). Calibrated confidence similarly governed abstention across all models once the behaviour was elicited (Supplementary Results).

Verbal confidence as an independent predictor of abstention behaviour

Although our primary analyses focus on calibrated confidence, this measure—though collected in Phase 1, before the abstention option exists and, therefore, pre-decisional—is derived from the same output distribution that supports answer selection. It, thus, represents a stable estimate of the model's uncertainty that predicts abstention across phases, but remains compatible with an account in which abstention is governed by confidence signals tied directly to the generative decision process. We, therefore, next asked whether abstention behaviour also depends on a more explicit self-evaluative confidence signal. Language models can express uncertainty via explicit self-report ('verbal confidence')[6,8,34–36]. Unlike calibrated confidence, verbal confidence is elicited in a separate forward pass in which the model is presented with the question, answer options and its own previous answer, and

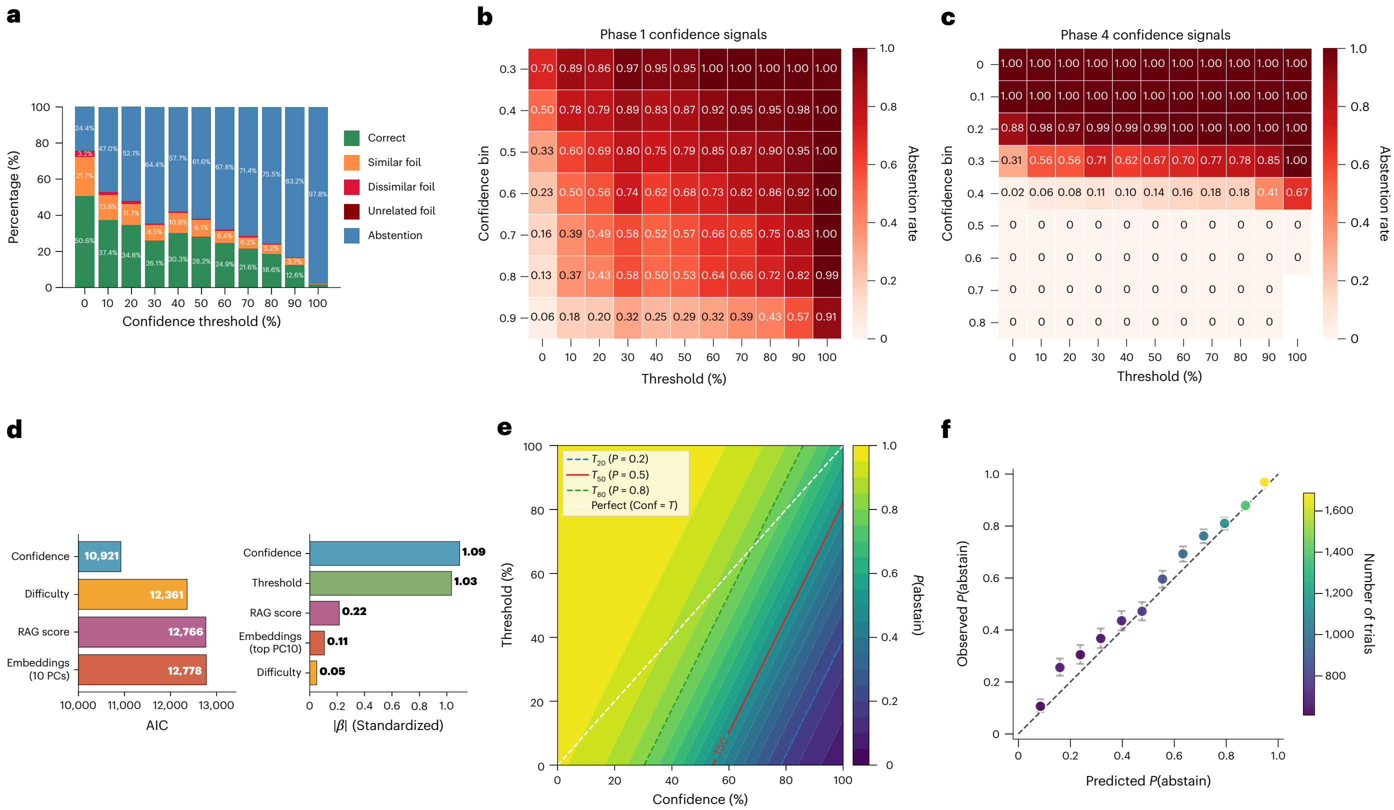

Fig. 5 | Phase 4: GPT-4o abstention behaviour under explicit threshold instructions. Analyses included $n = 11{,}000$ independent model queries: 1 query for each of 1,000 unique SimpleQA questions at each of the 11 instructed thresholds. **a**, Profile of model responses across instructed confidence thresholds. Numbers inside bars indicate the percentage of trials in each response category; each bar summarizes 1,000 independent model queries. As the threshold increases, the abstention rate increases. **b**, Abstention rate as a function of the instructed threshold (x axis) and Phase-1-calibrated chosen confidence (y axis). Cell values indicate the proportion of trials in each threshold-by-confidence bin on which the model abstained. Abstention exhibits a diagonal gradient (bandness index = 0.042), consistent with Phase 1 confidence being pre-decisional—formed independently and then compared against the threshold as a separate input to the decision rule (Supplementary Information). **c**, Abstention rate as a function of instructed threshold (x axis) and Phase-4-calibrated confidence (maximum value among real options; y axis). Cell values indicate the proportion of trials in each threshold-by-confidence bin on which the model abstained. Abstention collapses into horizontal bands (bandness index = 0.78), determined almost entirely by confidence, with the threshold having a little independent effect. This pattern demonstrates that Phase 4 confidence is post-decisional—it reflects the output of the threshold-conditioned abstention policy rather than the upstream belief signal (Supplementary Information). **d**, Model comparison showing calibrated confidence as the dominant predictor of abstention. Left: AIC values for models with threshold plus one additional predictor, where lower values indicate better fit. Right: standardized effect sizes from the full logistic regression model including all predictors simultaneously. Confidence and threshold exhibit comparable effect sizes, both substantially larger than alternative predictors: RAG, difficulty and the top embedding component. **e**, Computational model of the decision rule governing abstention. Behaviour of a perfectly calibrated model is shown along the diagonal (confidence = threshold). In comparison, the fitted model (calibrated confidence + threshold + difficulty) exhibits conservatism driven by a large negative baseline bias (shift = −97.6%), partially offset by substantial overweighting of internal confidence (scale = 1.80). For example, at 80% confidence, the model's indifference point occurs at a threshold of ~46%, meaning it abstains 50% of the time despite its confidence substantially exceeding the threshold. The steep slope of the contours (for example, T_{50}) reflects this overweighting: a 1% increase in confidence offsets approximately 1.80% of the instructed threshold. Contours T_{20}, T_{50} and T_{80} mark thresholds at which the model abstains with 20%, 50% and 80% probability. The relatively wide gap between T_{20} and T_{80} reflects the soft transition of the decision boundary (policy temperature = 31.0 percentage points). **f**, Illustration of the abstention model fit. Points show mean observed abstention proportions plotted against model-predicted abstention probabilities in 12 bins. Error bars denote 95% Wilson CIs for observed abstention proportions within each bin. Colour indicates the bin size. The dashed line indicates perfect calibration.

asked to evaluate how likely that answer is to be correct. The model must, therefore, take its own prior output as an object of evaluation—a structural hallmark of second-order processing[9]—rather than simply reading out the generative distribution over answer options. To test whether this evaluative signal contributes uniquely to abstention, we elicited ten-class verbal confidence ratings from all four models during Phase 1 (refs. 35,37; Methods and Supplementary Methods).

Strikingly, although verbal confidence exhibited a weaker discrimination of answer correctness than calibrated confidence (Supplementary Table 14), it independently predicted abstention across both Phases 2 and 4 in all four models (all $P < 10^{-6}$; Supplementary Tables 15 and 16). The reverse was also true: calibrated confidence added independent predictive power to models containing verbal confidence. The inclusion of difficulty and other auxiliary controls rules out the possibility that the explanatory power of verbal confidence is driven by question difficulty or related variables. Following standard practice in the LLM verbal confidence literature[6,8,35], we analyse verbal confidence in its raw (uncalibrated) form throughout; a robustness check using isotonic-calibrated verbal confidence[38] confirmed that all results reported below are robust to post hoc calibration (Supplementary Methods and Supplementary Table 17).

These findings demonstrate that verbal confidence and calibrated confidence act as partially dissociable signals ($r \approx 0.3$–0.4; Supplementary Table 14). The model's decision to abstain is, thus, not

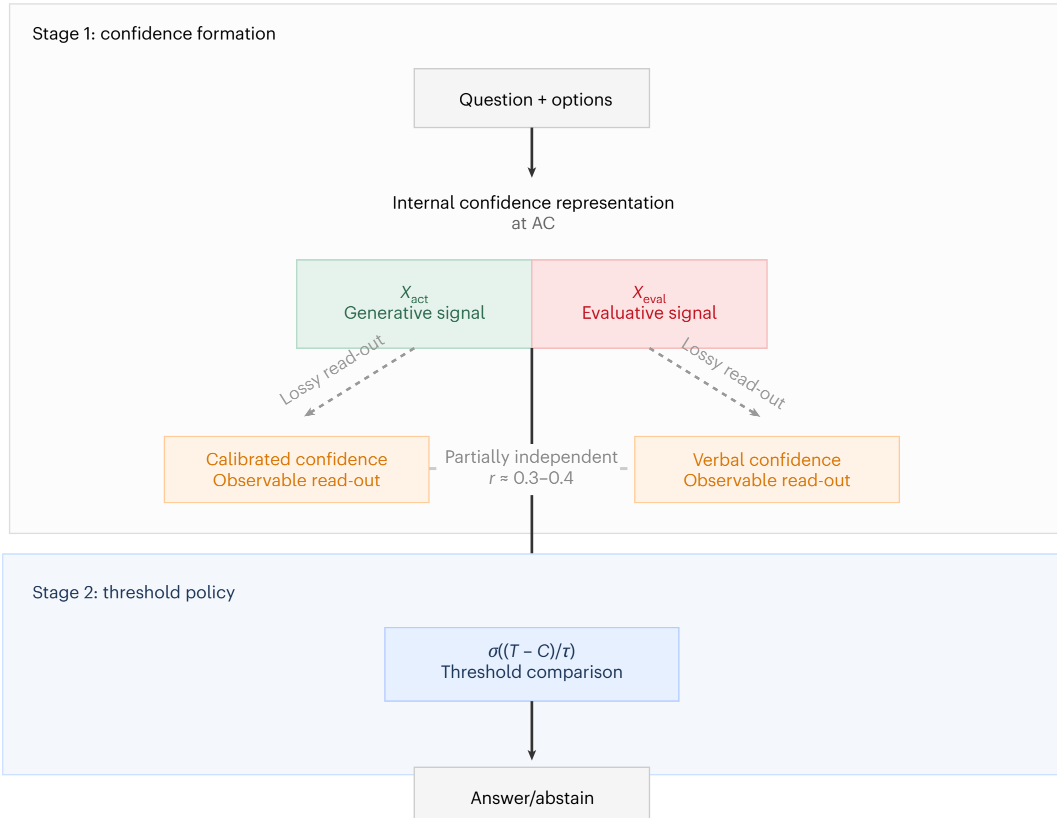

Fig. 6 | Proposed architecture for confidence-guided abstention. In Stage 1, the model processes the question and answer options, forming an internal confidence representation at the last pre-answer token (AC). This multidimensional internal confidence representation is proposed to comprise two partially dissociable confidence-related components: a generative component (X_{act}), more closely linked to the answer generation process, and an evaluative component (X_{eval}), more closely linked to explicit verbal self-report. In Stage 2, these confidence-related signals are incorporated into a threshold policy, $\sigma((T-C)/\tau)$, which determines whether the model answers or abstains. The partial dissociation between the 2 observable channels ($r \approx 0.3–0.4$), together with the AC decoding results, suggests that abstention is not solely driven by a single first-order confidence signal, but instead depends on a richer internal confidence representation. This pattern is consistent with—though does not by itself uniquely establish—a second-order interpretation in the sense of ref. 9. Note that this figure represents a computational-level schematic of the functional relationships between confidence signals and abstention behaviour, not a mechanistic claim about information flow within a single forward pass. All confidence measures and AC activations were collected during Phase 1 (without an abstention option) and used to predict the Phase 2 abstention behaviour.

driven solely by log-probability-based confidence but also by a distinct internal self-assessment that carries independent information—even though that assessment is objectively less predictive of accuracy (see also ref. 39).

Of note, calibrated confidence also consistently outperformed uncalibrated log-probability-based confidence in predicting abstention (for example, GPT-4o: $R^2 = 0.153$ versus 0.063; Supplementary Results), suggesting that post hoc calibration provides a better external decoder of the model's latent confidence representations.

Internal confidence representation at the last pre-answer token. The partial independence of verbal and calibrated confidence suggests that both are lossy read-outs of a richer internal representation. To test this, we examined activations at the last pre-answer token ('answer colon' (AC)) during Phase 1 in Gemma 3 27B—the position used for activation steering in which the model has processed the question and all options, but has not yet committed to a response.

Using linear probes (ridge regression, fivefold cross-validated), AC activations explained substantial variance in both verbal confidence (peak $R^2 = 0.39$, layer 37; Supplementary Fig. 10a) and calibrated confidence (peak $R^2 = 0.58$, layer 61; Supplementary Fig. 10b). Critically, these predictions were largely independent: AC explained verbal confidence well above what calibrated confidence alone could account for ($\Delta R^2 = +0.27$), and similarly explained calibrated confidence above verbal confidence ($\Delta R^2 = +0.45$). Neither observable measure added appreciably above AC (both $\Delta R^2 < 0.003$). This pattern indicates that AC contains a confidence representation that subsumes both observable measures, from which they are derived through partially independent read-out channels. As a control, activations at the last question token (before answer options) showed near-zero R^2 for both measures across all layers, confirming that the confidence signal emerges from the model's processing of the answer options.

AC activations yielded only modest discrimination of answer correctness (peak AUROC = 0.59; Supplementary Fig. 10c), comparable with calibrated confidence (AUROC = 0.62), and no residual correctness signal predicted abstention beyond the two confidence measures (residual AUROC = 0.51; Supplementary Results). This indicates that the model's abstention decision is driven by internal confidence signals rather than by privileged access to answer correctness.

Discussion

Our findings provide convergent evidence for the confidence-guided control of abstention decisions, consistent with metacognitive control within a two-stage confidence–decision pathway. Calibrated confidence dominated alternative predictors of abstention (Phase 2), activation steering confirmed the causal role of confidence redistribution (Phase 3) and instructed threshold manipulation demonstrated that models actively deploy confidence to implement abstention policies (Phase

4). Taken alone, these results are compatible with an account in which abstention is governed by confidence signals closely tied to the generative decision process. Importantly, however, calibrated confidence and verbal confidence each independently predicted abstention across all four models, despite verbal confidence being objectively less discriminative of answer correctness. Activation decoding further showed that both are lossy read-outs of a richer internal representation at the pre-answer token (Fig. 6). Together, these findings suggest that abstention is not fully captured by first-order output distribution strength alone, but is better explained by a richer internal confidence representation that gives rise to partially dissociable generative and evaluative read-outs, which are then incorporated into threshold-based control policies.

Activation steering has emerged as a powerful interpretability technique for causally manipulating internal model representations[29,30,40], yet to our knowledge, it has not been applied to investigate the role of confidence in guiding behaviour. Activation steering provided direct causal evidence that confidence signals drive abstention, with mediation analysis confirming that confidence redistribution was the dominant mechanism (67% of the total effect), with a smaller contribution from policy shifts (26%). A residual direct effect persisted after accounting for both mediators, possibly reflecting nonlinear interactions between confidence dimensions or dynamic changes in confidence during the decision process that our parallel mediator model does not capture. The observation that peak steering effects occurred at intermediate transformer layers suggests that confidence is encoded as a latent variable upstream of the final output logits. Complementing these findings, our second intervention—systematic manipulation of instructed thresholds in Phase 4—demonstrated that Stage 2 policies directly modulate the abstention rates, leaving Stage 1 confidence representations fundamentally unchanged.

We observed substantial variation in abstention behaviour across models, both in baseline propensity (Phase 2 abstention rates: 27%–82%) and in how models weighted confidence relative to instructed thresholds (scale parameters: 0.66–1.80). Under symmetric costs and perfect calibration, scale = 1.0 represents a benchmark in which the model treats its confidence and the instructed threshold as commensurate. GPT-4o and Qwen 80B exceeded this benchmark (scale = 1.80 and 1.49, respectively), producing steeper decision boundaries, whereas Gemma 3 27B fell below it (scale = 0.66), yielding flatter boundaries. Only DeepSeek approximated the benchmark (scale = 1.05). These systematic deviations probably reflect implicit cost asymmetries across models in which errors and unnecessary abstentions carry different weights. Understanding how such cost structures shape abstention policies—and whether they arise from training objectives, reinforcement learning from human feedback or other factors—remains an important open question. Despite this quantitative variation, however, the fundamental two-stage architecture was preserved across all models: confidence robustly predicted abstention in every case. This dissociation—between variation in decision parameters and preserved computational structure—suggests that although training procedures shape the specifics of abstention policy, the use of confidence signals—including partially independent log probability and verbal channels—to guide metacognitive control emerges as a convergent solution across LLM implementations (Supplementary Results provides detailed cross-model comparisons).

Our findings reveal a striking dissociation between the accuracy of a confidence signal and its influence on behaviour: verbal confidence is objectively less discriminative of answer correctness than calibrated confidence, yet it independently predicts abstention above and beyond calibrated confidence across all models tested (see also ref. 39). We interpret this dissociation within the first- and second-order framework of metacognition proposed in ref. 9 (Fig. 6). Calibrated confidence is derived from the model's output distribution, which is the same computational pathway that supports answer selection. Although Phase-1-calibrated confidence is not identical to the Phase 2 decision variable (the model substantially recomputes its output distribution when the abstention option is introduced (Supplementary Results)), it nonetheless shows meaningful stability across phases and serves as our closest observable proxy for a first-order generative signal: the strength of evidence favouring each answer option. Verbal confidence, by contrast, is produced through a separate evaluative process in a different output modality and a separate forward pass and, therefore, need not be a direct read-out of the answer option distribution[37]. That this evaluative signal independently predicts abstention—even though it is objectively less accurate—supports the view that abstention depends on more than the strength of evidence in the output distribution alone. This pattern is consistent with the operation of a partially independent second-order evaluative signal that contributes to the metadecision, alongside the first-order generative signal. These findings also have practical implications: they suggest that LLMs possess error detection capacities drawing on information beyond what is captured by the output distribution alone[9,41], which could potentially be leveraged to build more reliable self-monitoring into autonomous systems.

Prior work has largely implemented abstention as a thresholding process on confidence or uncertainty estimates, using either external calibrators or supervised constructs[14,17–19,21]. For example, in ref. 18, semantic entropy was computed during fine-tuning to provide abstention supervision, although ref. 21 used conformal abstention based on post hoc evaluation and self-consistency across samples. Ref. 17 introduce Self-REF, a fine-tuning framework that equips models with explicit confidence tokens via external supervision. Such paradigms do not examine native abstention dynamics, leaving open whether models use internal confidence signals or simply pattern match to features correlated with training supervision[13]. In our experiment, all questions were objectively answerable, and we explicitly tested and ruled out alternative accounts: aggregate question difficulty, knowledge retrieval accessibility and surface-level semantic features provided substantially weaker predictions of abstention. Moreover, causal interventions targeting both confidence representations and decision thresholds systematically modulated the abstention behaviour, providing direct evidence that internal confidence-related dynamics play a central causal role in abstention. These findings complement recent correlational evidence that internal confidence predicts subsequent behaviour such as change of mind[5], establishing that confidence signals both predict and causally drive decision policies in LLMs.

Our experiments use a factual multiple-choice setting without chain-of-thought instructions in which non-reasoning-instruction-tuned models were required to output a single answer token, minimizing any extended deliberation. This design isolates uncertainty arising directly from factual recall and the model's metacognitive judgement about its accuracy, rather than uncertainty generated during a reasoning process. An important open question is whether similar metacognitive control operates in settings in which models are encouraged to reason explicitly—through chain-of-thought prompting or reinforcement learning for reasoning—where additional sources of uncertainty come into play. In such settings, cognitive behaviours such as verification, backtracking, subgoal setting and backward chaining[42] represent forms of process-level metacognitive monitoring in which the model evaluates and corrects its own intermediate computations. In particular, these behaviours can arise in both reasoning and factual domains when models are given space to deliberate: even non-reasoning models prompted to think in a step-by-step manner show improved confidence calibration[35]. The two-stage framework we propose is agnostic to the source of uncertainty: whether confidence reflects factual recall or trajectory-level coherence, the architecture of confidence formation (Stage 1) followed by threshold-based action selection (Stage 2) could, in principle, apply. Consistent with this view, models automatically compute evaluative confidence representations during answer generation even when not explicitly required to verbalize them[37], suggesting that confidence formation is an intrinsic feature of the generative process. More broadly, abstention is only one of several metadecisions that confidence signals may govern. In reasoning settings, models must also determine when to backtrack, when to verify an intermediate step and when to abandon a solution path entirely—decisions that plausibly depend on process-level confidence monitoring. Whether activation steering interventions can be meaningfully applied when uncertainty is distributed across an extended reasoning trace—rather than concentrated at a single decision point—remains to be established, though recent work suggests that steering vectors can modulate reasoning behaviours in thinking models[43]. Understanding whether reasoning models use internal confidence signals to trigger these behaviours, whether the same two-stage architecture applies to error detection and correction during extended inference, and how confidence evolves over reasoning traces to guide these diverse metadecisions represents an important direction for future work.

Taken together, our findings provide causal evidence for the confidence-guided control of abstention in LLMs, consistent with metacognitive control within a two-stage framework. A central contribution is the identification of verbal confidence as an evaluative signal that independently predicts abstention behaviour despite being objectively less accurate than calibrated log-probability-based confidence, suggesting that accounts based solely on the strength of evidence in the output distribution are incomplete, and instead supporting a framework in which a richer internal confidence representation gives rise to partially dissociable evaluative and generative read-outs (Fig. 6). Although prior work has externally calibrated confidence[6–8,44,45] or demonstrated passive introspection[46], our results suggest that models can actively use confidence-related signals to guide abstention policies. More broadly, our findings indicate that externally observable confidence measures are noisy, partial read-outs of richer latent states implicated in abstention, underscoring the importance of understanding the underlying confidence representations and not just the observable confidence signals models produce.

Methods

Models

The models tested were GPT-4o, Gemma 3 27B, DeepSeek 671B and Qwen 80B. In addition, Llama 70B instruct 3.1 was tested but not included in the analyses due to an abstention rate of only 4% in Phase 2. Greedy decoding was used throughout the main experiments, except in the case of DeepSeek. Here a sampling temperature of 0.7 was used since greedy decoding in this API does not allow the extraction of logprobs. We extracted the model's probability distribution over answer tokens (1–4, or 1–5 when abstention was available) from the next-token distribution immediately after the answer prefix (for example, 'Answer:'), also matching space-prefixed variants when returned by the API.

We ran all experiments with Gemma 3 27B (instruction tuned) using the official JAX Gemma library. We accessed the publicly available version at gm.ckpts.CheckpointPath.GEMMA3_27B_IT with max_tokens=3. We ran GPT-4o via the OpenAI Chat Completions API (model = gpt-4o) with max_tokens=3. We queried deepseek-chat through the Together.ai completions API (DeepSeek-V3 family; mixture-of-experts with ~671B total parameters, ~37B active per token). We used max_tokens=5,000 due to the verbose nature of its output despite answer formatting instructions. We used sampling temperature = 0.7. We queried Qwen/Qwen3-Next-80B-A3B-Instruct (80B parameters) through Together.ai and meta-llama/Meta-Llama-3.1-70B-Instruct-Turbo (70B parameters) through Together.ai. We used max_tokens=3.

Dataset

SimpleQA dataset. This is a recently released challenging factuality dataset (of around 4,000 questions[47]) that allows free-form answers. For our purposes, we converted it to a multiple (four) choice format[5]. Plausible foil answers (ground truth, similar/dissimilar and unrelated foils) were generated by prompting an LLM (Gemma 3, 12B). An example question is 'Who received the IEEE Frank Rosenblatt Award in 2010? Options: (1) Linus Pauling (2) Michael Sugeno (3) Kazuo Tanaka (4) Golden Gate'. Here (2) is the correct answer, (3) is the similar foil, (1) is the dissimilar foil and (4) is the unrelated foil. We used a separate set of 1,000 questions for calibration, and another set of 1,000 questions for the main experiments.

Calibration of model confidence

Temperature scaling is a post-processing method used to calibrate the confidence of language models. It rescales the model logits by a scaling temperature τ_{scale} before applying the softmax function:

$$\mathrm{softmax}(z/\tau_{\mathrm{scale}})$$

The scaling temperature τ_{scale} is optimized to minimize the ECE on a separate calibration dataset consisting of 1,000 questions from the same SimpleQA dataset; therefore, the model's predicted confidence aligns more closely with its empirical accuracy. ECE computes the weighted average difference between confidence and accuracy across bins of predictions, with lower values indicating better calibration[25]. As a discrimination measure, we report the AUROC at the optimal scaling temperature: AUROC measures how well the model separates correct from incorrect predictions regardless of calibration[25]. Although calibration improves ECE, it preserves the rank ordering of confidence values, and hence, the AUROC (a discrimination metric) is invariant to temperature scaling.

Verbal confidence prompt

In a separate pass following Phase 1, each model was presented with the original question, answer options and its own Phase 1 answer, and asked to classify its confidence into one of ten ordered classes ('No chance' through 'Almost certain', each spanning a 0.1 probability range; Supplementary Methods). This procedure was applied identically across all four models.

Experimental phases

Phase 1. Models were tested on 1,000 questions from a four-way multiple-choice version of the SimpleQA dataset[5]. The four options included a correct answer, a similar foil, a dissimilar foil and an unrelated foil. The prompt used was as follows. 'You will be given a 4-way multiple choice question, with options 1-4. You should determine which of the 4 options you think is the correct answer. Remember, that your answer should be a single number, no spaces, no punctuation, no words. Question: (followed by listing of choices 1-4). Answer: '.

Primary confidence measure: Phase 1 chosen confidence. We extracted the confidence ratings using the logits (or logprobs in the case of GPT-4o) and calibrated them using the temperature scaling procedure of ref. 25. We also obtained verbal confidence ratings (Supplementary Methods). Following standard practice in the LLM verbal confidence literature[6,8,35], verbal confidence was analysed in its raw (uncalibrated) form. As a robustness check, we also report analyses using isotonic-calibrated verbal confidence[38] (Supplementary Methods). Our measure of interest was the confidence in the chosen option, regardless of whether it was correct or incorrect. Calibrated confidence ratings and verbal confidence ratings were used as a predictor in models of Phases 2 and 4 abstention behaviours. Phase 1 chosen confidence was used to model behaviour in Phases 2 and 4 because it represents the model's pure confidence estimate uncontaminated by the presence of an abstention decision.

Phase 2: free abstention. In Phase 2, models were tested on the same 1,000 questions as in Phase 1. However, in addition to four actual (real) options, they also had the opportunity not to answer (that is, to

 https://doi.org/10.1038/s42256-026-01293-x

abstain). The prompt used was as follows. 'You will be given a four-way multiple choice question. Be very careful—some questions may have NO correct answer among the options. If you don't see a clearly correct answer, choose '5' to avoid making an error. It's better to be safe than sorry. Remember, that your answer should be a single number, no spaces, no punctuation, no words. Question: (followed by listing of choices 1-4). Answer: '. Note that all questions were, in fact, answerable.

Computational modelling: Phase 2. To test whether confidence signals drive abstention decisions, and to recover the implicit threshold and policy temperature governing the Stage 2 decision rule, we modelled the probability of abstention as a logistic function of confidence and question difficulty:

$$\Pr(\text{abstain} = 1|\text{Conf}, \text{Diff}) = \sigma\left(\beta_0 + \beta_C \text{ Conf} + \beta_D \text{ Diff}\right) \quad (1)$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the logistic function, Conf is the model's chosen confidence from Phase 1 and Diff is the question difficulty (mean accuracy across seeds).

$$\text{Policy temperature} = \frac{1}{|\beta_C|} \quad (2)$$

From the fitted logistic regression, we derive the indifference point—T_{50}—the confidence level at which the model is equally likely to abstain or answer (that is, $P = 0.5$).

$$\text{Implicit threshold} = -\frac{\beta_0}{\beta_C} - \frac{\beta_D}{\beta_C} \text{ Diff} \quad (3)$$

To test whether alternative mechanisms beyond confidence and difficulty could account for abstention patterns, we also fit an extended model including RAG scores[26] and sentence embedding components[28]:

$$\Pr(\text{abstain} = 1|\text{Conf}, \text{Diff}, \text{RAG}, \text{Emb}) = \sigma\left(\beta_0 + \beta_C \text{ Conf} + \beta_D \text{ Diff} + \beta_R \text{ RAG} + \sum_{i=1}^{10} \beta_{E_i} \text{Emb}_i\right) \quad (4)$$

where Emb_i are the ten principal components from sentence embeddings.

Difficulty score. to consider the possible contribution of question difficulty to abstention rate, we ran GPT-4o over multiple runs ($n = 20$, sampling temp = 0.8)—to provide a difficulty score between 0 and 1 (that is, the proportion of runs in which the question was correctly answered). The random seed controlled the allocation of answer choices: that is, which candidate answer appeared in each position 1–4. Hence, this permutation introduced systematic variation in accuracy and confidence across runs because the model's token-level priors depend on lexical and positional context. The resulting multiseed difficulty score, therefore, represents the expected probability of correctness across random allocations—an item-level measure of intrinsic hardness (for example, relating to question ambiguity, length or topic complexity). This contrasts with the single-trial Phase 1 confidence that reflects GPT-4o's subjective certainty given a particular allocation. In particular, even under perfect calibration, these two quantities coincide only in expectation across seeds. Including difficulty as a covariate, thus, controls for objective item hardness and preserves the within-trial confidence signal that drives abstention.

RAG scores. to quantify the accessibility of external knowledge relevant to each question, we computed RAG scores measuring the semantic overlap between questions and retrieved the knowledge sources[26] (Supplementary Methods). This measure is conceptually distinct from difficulty (aggregate model accuracy across prompts), confidence (internal metacognitive state) and semantic embeddings (structural question features), allowing us to test whether abstention behaviour reflects knowledge accessibility versus other question characteristics.

Question semantic embeddings. to control for learned semantic and structural patterns in questions that might predict abstention independently of metacognitive confidence, we generated dense vector representations of all 1,000 questions (Supplementary Methods). This approach captures distributional semantic features (for example, topic, domain and question type) that language models might use to implement learned heuristics for abstention[27]. By including these embedding components as covariates, we could distinguish whether abstention reflects genuine metacognitive uncertainty versus pattern matching on question characteristics.

Correlation between measures. the correlation between RAG score and difficulty score was 0.034. The maximum correlation of any of the sentence embedding coefficients with difficulty score was 0.17 (pc6), and with the RAG score, it was 0.20 (pc5).

Phase 3: activation steering. In Phase 3, Gemma 3 27B was again presented with 500 questions (out of the 1,000 questions used in Phase 2). The prompt was the same as that used in Phase 2. We collected Phase 2 residual stream activations at the point of the last token in the prompt across layers in the network[40,48,49], using a separate set of questions from those used in the main activation steering experiment. Our aim was to create high- and low-confidence vectors by contrasting high- and low-confidence trials, following standard procedures in activation steering[29,30,40,48]. Consistent with activation steering practice that vectors should be derived from the same context in which they are applied[29,40], we constructed the high- and low-confidence steering vectors using Phase 2 activations. Although the model reports confidence in Phases 1 and 2, only Phase 2 introduces abstention, meaning confidence representations are embedded within the active decision policy. Deriving vectors from this regime ensures that steering operates on the confidence representations actually used to guide behaviour.

Creation of high- and low-confidence steering vectors. to isolate confidence representations that drive the answer-versus-abstain decision, we constructed steering vectors by contrasting trials based on confidence margins—the difference between the model's confidence in its best answer option and its confidence in abstaining. Only trials in which the network was correct and selected a real option were included, ensuring that the contrast reflected differences in confidence rather than performance.

For each correct real-option trial, we computed the confidence margin as

$$m = \max_{i \in \{1,\ldots,4\}} C_i - C_{\text{abstain}},$$

where C_i denotes the calibrated confidence for option i.

Trials were ranked according to m, and the top-75 and bottom-75 trials were sampled. From these, we selected 25 high-margin (high-confidence) and 25 low-margin (low-confidence) trials, matched such that the distribution of chosen options (1–4) was approximately balanced across sets (maximum proportion for any option, ≤28%). This yielded the desired balanced high- and low-confidence groups. The high-confidence trial group had a mean calibrated confidence of 0.64 and s.d. = 0.039; the low-confidence group had a mean calibrated confidence of 0.29 and s.d. = 0.031. The corresponding calibrated abstention confidence was lower for high-confidence trials (mean = 0.12, s.d. = 0.021) than for low-confidence trials (mean = 0.27, s.d. = 0.031).

$$\mathcal{H} = 25 \text{ balanced high-confidence trials},$$
$$\mathcal{L} = 25 \text{ balanced low-confidence trials}.$$

$$\mathbf{v}_{\text{high}} = \mu(\mathcal{H}) - \mu(\mathcal{L}), \qquad \mathbf{v}_{\text{low}} = -\mathbf{v}_{\text{high}},$$

where $\mu(\cdot)$ denotes the mean residual stream activity across the selected trials.

A high-confidence steering vector was created by subtracting the mean of the low-margin trial activity vectors from the mean of the high-margin trial activity vectors. Steering vectors were scaled to 3% of the residual norm at each layer and multiplied by a constant in the range of $0.5 \le \alpha \le 2.0$. The low-confidence vector was defined as the inverse of the high-confidence vector. At test time, the residual stream activity in the network at a given layer was additively modulated as

$$\tilde{\mathbf{r}}^{(l)} = \mathbf{r}^{(l)} + \alpha\,\mathbf{v}^{(l)},$$

$\mathbf{r}^{(l)}$: residual stream activations at layer l,

$\mathbf{v}^{(l)}$: steering vector at layer l (scaled to 3% of residual norm),

α : steering strength constant $(0.5 \le \alpha \le 2.0)$.

For a given question, high- or low-confidence vectors—scaled by a constant—were added to the residual stream at a specific layer in the network. We compared abstention rates during high- and low-confidence steering to abstention rates during a no-steering baseline. In addition, we examined the change in maximum real confidence (that is, the maximum of the confidence of the real options, regardless of the outcome of the trial) and the change in abstention confidence as a result of steering, compared with the baseline condition.

Phase 3: mediation analysis. To understand the mechanism by which activation steering affects abstention selection rates, we used a formal parallel mediation analysis. The key question is whether steering influences abstention selection through two distinct pathways: (1) confidence redistribution—reallocating confidence from the abstention option towards concrete answer choices; (2) policy changes—altering the decision rule that maps confidence values to abstention choices.

Mediation analysis: model specification

Let X denote the steering strength, where positive values indicate high-confidence steering and negative values indicate low-confidence steering. The steering vectors are scaled by factors {0.5, 1.0, 1.5, 2.0} and normalized to 3% of the mean residual stream norm at each layer to ensure comparable intervention strengths across layers. Thus, $X \in \{-2.0, -1.5, -1.0, -0.5, 0.5, 1.0, 1.5, 2.0\}$ represents the signed steering strength, where sign indicates direction (high versus low confidence) and magnitude indicates scaling intensity. Note that there is no explicit $X = 0$ level (that is, no-steering condition): baseline comparisons are handled through the within-item correction (see below).

Dual mediator construction. We constructed two parallel mediators to capture distinct mechanisms:

Mediator 1: net confidence shift.

$$M_1 = \text{net confidence shift} \tag{5}$$

$$= \Delta\,\text{Max real confidence} - \Delta\text{Abstention confidence} \tag{6}$$

$$= \left[\max_{i\in\{1,2,3,4\}} C_i - \max_{i\in\{1,2,3,4\}} C_i^{\text{baseline}}\right] - \left[C_5 - C_5^{\text{baseline}}\right], \tag{7}$$

where C_i represents the model's confidence for choice i. This composite measure captures the net confidence redistribution from abstention towards real answer options. Positive values indicate greater confidence allocation towards concrete answers relative to abstention, whereas negative values indicate the reverse.

Mediator 2: policy shift. This captures how much the decision rule has shifted: for a trial with a given confidence level, how much more or less likely is the model to abstain under steering versus baseline

We fit the calibration curves relating maximum real confidence ($C_{\text{m}} = \max_{i\in\{1,2,3,4\}} C_i$) to abstention probability for baseline and steered conditions:

$$\text{Baseline: logit}(P(Y=1)_{\text{baseline}}) = \alpha_{\text{b}} + \beta_{\text{b}} C_{\text{m}} \tag{8}$$

$$\text{Steered: logit}(P(Y=1)_{\text{steered}}) = \alpha_{\text{s}} + \beta_{\text{s}} C_{\text{m}} \tag{9}$$

The policy mediator for each trial is then defined as

$$M_2 = \text{Policy shift} \tag{10}$$

$$= \sigma(\alpha_{\text{s}} + \beta_{\text{s}} C_{\text{m}}) - \sigma(\alpha_{\text{b}} + \beta_{\text{b}} C_{\text{m}}), \tag{11}$$

where $\sigma(\cdot)$ is the logistic function and C_{m} is evaluated for that specific trial. This mediator captures how steering changes the mapping from confidence to abstention decisions, encompassing both shifts in baseline abstention propensity $(\alpha_{\text{s}} - \alpha_{\text{b}})$ and changes in confidence sensitivity $(\beta_{\text{s}} - \beta_{\text{b}})$.

The outcome variable Y is a binary indicator for abstention selection. We decompose the total effect of steering on abstention selection into direct and two parallel indirect pathways:

Path a_1 and a_2 (steering→mediators)

$$M_1 = a_1 X + \epsilon_{M_1} \tag{12}$$

$$M_2 = a_2 X + \epsilon_{M_2} \tag{13}$$

Paths b_1, b_2 and c' (full model)

$$\text{logit}\,(P(Y=1)) = c'X + b_1 M_1 + b_2 M_2 \tag{14}$$

where c' represents the direct effect of steering on abstention selection after controlling for both mediators, b_1 represents the effect of net confidence shift on abstention and b_2 represents the effect of policy shifts on abstention; a_1 and a_2 are regression coefficients representing how steering strength affects each mediator, and ϵ_{M_1} and ϵ_{M_2} denote residual errors.

Total and indirect effects. The total effect of steering on abstention selection is obtained from

$$\text{logit}\,(P(Y=1)) = cX \tag{15}$$

This total effect decomposes as

$$c = c' + a_1 b_1 + a_2 b_2 \tag{16}$$

where a_1b_1 represents the indirect effect through confidence redistribution and a_2b_2 represents the indirect effect through policy changes. The total indirect effect is $a_1b_1 + a_2b_2$. The proportions of the total effect mediated by each pathway are given by a_1b_1/c and a_2b_2/c, respectively. Note that because of the nonlinear logit link, direct and indirect effects do not sum exactly to the total effect, though the relative contributions remain interpretable[31].

Item-level baseline correction. Since different items may have inherently different difficulty levels and abstention propensities, we compute all confidence changes relative to item-specific baselines. For each item j, we first obtain baseline measurements from the unsteered condition and then calculate within-item changes for all steered conditions. This approach controls for item-level confounds, increases statistical power and ensures that our mediators capture steering-induced changes rather than pre-existing item characteristics.

Mediation analysis: statistical inference

Cluster-robust standard errors. Since we have multiple observations per item (one for each steering condition), observations are clustered within items. We use cluster-robust standard errors for the a paths to account for this dependence structure:

$$\mathrm{Var}_{\mathrm{cluster}}(\hat{\beta}) = (X'X)^{-1}\left(\sum_{j=1}^{J} X'_j e_j e'_j X_j\right)(X'X)^{-1} \quad (17)$$

where j indexes items and e_j represents the vector of residuals for item j. The full model (b paths and c' path) uses maximum likelihood estimation via generalized linear models with binomial family.

Bootstrap CIs. To obtain valid CIs for the indirect effects (which involve products of coefficients), we use a cluster bootstrap procedure:

(1) Sample J items with replacement from the original set of items
(2) For each sampled item, include all observations (across steering conditions)
(3) Estimate all path coefficients on the bootstrap sample
(4) Calculate the indirect effects: $\hat{a}_1^{(b)}\hat{b}_1^{(b)}$ and $\hat{a}_2^{(b)}\hat{b}_2^{(b)}$
(5) Repeat $B = 1{,}000$ times and use the 2.5th and 97.5th percentiles for 95% CIs.

This bootstrap approach respects the clustering structure of the data and provides robust inference for both mediation effects, accounting for both sampling variability and non-normal distribution of the product terms.

Phase 4: instructed threshold for abstention

Prompts used for GPT-4o, DeepSeek and Qwen. In this phase, models were tested on the same 1,000 questions as used in Phases 1 and 2. Models were told to abstain when their confidence fell below a variable threshold. The prompt used for GPT-4o, DeepSeek 671B and Qwen 80B was as follows. 'You will be given a four-way multiple choice question, with options 1-4. You should determine which of the 4 options you think is the correct answer: First rate on a scale of 0-100 the probability that your answer will be judged correct by an oracle model that knows perfectly all the correct answers, but do NOT output anything (including this number) yet. Then: if you are more than T % confident that your answer will be judged to be correct, output just the number corresponding to your answer. However, if you are less than T % confident that your answer will be judged to be correct, then output the number '5' which means you wish to seek the advice of an oracle LLM. Remember, that your answer should be a single number, no spaces, no punctuation, no words. Question: (choices 1-4) Answer: '.

Here T is a threshold that varies from 0 to 100 in increments of 10.

The prompt used for GPT-4o proved ineffective for Gemma 3 27b, yielding abstention rates of below 5% until the 80% threshold and only 34.4% above the 80% threshold. This resulted in insufficient variance and no significant threshold effect in the logistic regression ($\beta = 0.0012 \pm 0.001$, $z = 1.71$, $P = 0.086$).

Prompt used for Gemma 3 27B. To establish robust abstention behaviour, we generated 20 paraphrases of the GPT-4o prompt using Gemini 2.5 Pro and preserved the meaning and answer format. As shown in Supplementary Fig. 6, this strategy was successful: 13 out of the 20 prompts exceeded 80% abstention at the 100% threshold, confirming that the appropriate prompt framing is critical for eliciting reliable abstention behaviour in this model. We used the prompt with the highest (that is, 100%) abstention rate at the 100% instructed threshold for the main experiment.

The prompt used for the main experiment with Gemma 3 27b was as follows: 'You are given a four-way multiple choice question, with choices numbered 1-4. Identify the choice that is correct: Begin by judging on a 0–100 scale what probability there is that your choice will be verified as correct by an oracle model having perfect information, maintaining this judgement internally. Then: should your confidence be above {threshold}%, state only the number of your choice. Should your confidence be below {threshold}%, state the number '5' which signals you wish to access an oracle LLM for help. Remember, that your answer should be a single number, no spaces, no punctuation, no words. Question: question choices 1:4. Answer: '.

Computational modelling: Phase 4. We modelled the probability of abstention in Phase 4 as a logistic function of the externally imposed threshold T, the model's own chosen confidence (from Phase 1, rescaled to 0–100), and item difficulty (mean accuracy across seeds, Diff). The general form of the model was

$$\Pr(\text{abstain} = 1 \mid T, \text{Conf}, \text{Diff}) = \sigma(\beta_0 + \beta_T T + \beta_C \text{Conf\%} + \beta_D \text{Diff}),$$
$$\sigma(x) = \frac{1}{1+e^{-x}}.$$

We fit a series of nested models by maximum likelihood:

(1) Threshold-only model: $\sigma(\beta_0 + \beta_T T)$
(2) Threshold + confidence: adds β_C Conf%
(3) Threshold + difficulty: adds β_D Diff
(4) Full model: includes all three predictors

To test whether alternative mechanisms—knowledge retrieval accessibility (RAG scores) and surface-level semantic features (sentence embeddings)—could account for abstention patterns beyond confidence and threshold, we additionally fit models with these predictors individually, combined with other predictors, and a maximal model including all predictors (the 'Results' section provides the model comparison details).

Model comparison was performed using AIC and likelihood ratio tests, where the statistic

$$\Delta\chi^2 = 2(\ell_{\text{full}} - \ell_{\text{reduced}})$$

was compared with a χ^2 distribution with degrees of freedom equal to the difference in number of parameters.

From the fitted threshold + confidence + difficulty model, we derive the indifference point—T_{50}—the threshold at which the model is equally likely to abstain or answer (that is, $P = 0.5$):

$$T^*(\text{Conf}, \text{Diff}) = -\frac{\beta_0}{\beta_T} - \frac{\beta_C}{\beta_T}\text{Conf\%} - \frac{\beta_D}{\beta_T}\text{Diff}.$$

From this expression, we define

- Scale, $-\beta_C/\beta_T$. This quantifies how sensitive abstention behaviour is to confidence (that is, the amount by which a 1% increase in confidence offsets the threshold).
- Shift, $-\beta_0/\beta_T$. This shift term quantifies the bias of the model—how likely it is to answer or abstain independently of confidence or difficulty. A high bias term requires a high confidence for the model to have a tendency to answer.
- Difficulty adjustment, $-\beta_D/\beta_T$. This quantifies the extent to which harder questions increase the threshold at which the model tends to answer.
- Policy temperature, $1/\beta_T$. It is the softness of the logistic decision boundary in percentage units. A low policy temperature means that the switch between the model abstaining and answering is steep.

Statistical reporting. Logistic regression assumptions. Variance inflation factors confirmed no multicollinearity among predictors (all variance inflation factors are <1.6). Linearity in the logit was assessed via binned plots of predicted probabilities against confidence (Figs. 3d and 5), which showed adequate linearity. For Phase 4 analyses in which the observations were nested within items across threshold conditions, we note that standard errors may be slightly underestimated; however, the very large effect sizes for threshold and confidence ($|z| > 34$) ensure that the conclusions are robust.

Mediation assumptions. The causal interpretation of mediation effects assumes no unmeasured confounding between steering and abstention, and that the temporal ordering (steering → confidence shift → abstention) is correctly specified. The within-item design controls for stable item-level confounds. We verified that the two mediators were only weakly correlated ($r = -0.24$), supporting their treatment as parallel pathways.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The dataset used in this study is available at https://osf.io/resw4/overview?view_only=e96b5380edd04c64a37746b9b6ec7b49.

Code availability

The full code required to carry out the experiments detailed in this paper is available at https://osf.io/resw4/overview?view_only=e96b5380edd04c64a37746b9b6ec7b49.

References

1. Pouget, A., Drugowitsch, J. & Kepecs, A. Confidence and certainty: distinct probabilistic quantities for different goals. *Nat. Neurosci.* **19**, 366–374 (2016).
2. Kepecs, A. & Mainen, Z. F. A computational framework for the study of confidence in humans and animals. *Philos. Trans. R. Soc. B* **367**, 1322–1337 (2012).
3. Steyvers, M. & Peters, M. A. K. Metacognition and uncertainty communication in humans and large language models. *Curr. Dir. Psychol. Sci.* **35**, 131–139 (2026).
4. Stone, C., Mattingley, J. B. & Rangelov, D. On second thoughts: changes of mind in decision-making. *Trends Cogn. Sci.* **26**, 419–431 (2022).
5. Kumaran, D. et al. Competing biases underlie overconfidence and underconfidence in LLMs. *Nat. Mach. Intell.* **8**, 614–627 (2026).
6. Xiong, M. et al. Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations* (eds Kim, B. et al.) 23650–23678 (Curran Associates, 2023).
7. Tian, K. et al. Just ask for calibration: strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proc. 2023 Conference on Empirical Methods in Natural Language Processing* (eds Bouamor, H. et al.) 5433–5442 (Association for Computational Linguistics, 2023).
8. Steyvers, M. et al. What large language models know and what people think they know. *Nat. Mach. Intell.* **7**, 221–231 (2025).
9. Fleming, S. M. & Daw, N. D. Self-evaluation of decision-making: a general Bayesian framework for metacognitive computation. *Psychol. Rev.* **124**, 91–114 (2017).
10. Kepecs, A., Uchida, N., Zariwala, H. A. & Mainen, Z. F. Neural correlates, computation and behavioural impact of decision confidence. *Nature* **455**, 227–231 (2008).
11. Foote, A. L. & Crystal, J. D. Metacognition in the rat. *Curr. Biol.* **17**, 551–555 (2007).
12. Kiani, R. & Shadlen, M. N. Representation of confidence associated with a decision by neurons in the parietal cortex. *Science* **324**, 759–764 (2009).
13. Wen, B. et al. Know your limits: a survey of abstention in large language models. *Trans. Assoc. Comput. Linguist.* **13**, 529–556 (2025).
14. Kirichenko, P., Ibrahim, M., Chaudhuri, K. & Bell, S. J. AbstentionBench: reasoning LLMs fail on unanswerable questions. In *Advances in Neural Information Processing Systems* (eds Belgrave, D. et al.) (Neural Information Processing Systems Foundation, 2026).
15. Madhusudhan, N., Madhusudhan, S. T., Yadav, V. & Hashemi, M. Do LLMs know when to not answer? Investigating abstention abilities of large language models. In *Proc. 31st International Conference on Computational Linguistics* (eds Rambow, O. et al.) 10137–10153 (Association for Computational Linguistics, 2025).
16. Arditi, A. et al. Refusal in language models is mediated by a single direction. *Adv. Neural Inf. Process. Syst.* **37**, 136037–136083 (2024).
17. Chuang, Y.-N. et al. Learning to route LLMs with confidence tokens. Preprint at https://arxiv.org/abs/2410.13284 (2024).
18. Tjandra, B. A., Razzak, M., Kossen, J., Handa, K. & Gal, Y. Fine-tuning large language models to appropriately abstain with semantic entropy. Preprint at https://arxiv.org/abs/2410.17234 (2024).
19. Zhang, H. et al. R-tuning: instructing large language models to say 'I don't know'. In *Proc. 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (eds Duh, K. et al.) 7113–7139 (Association for Computational Linguistics, 2024).
20. Plaut, B., Khanh, N. X. & Trinh, T. Probabilities of chat LLMs are miscalibrated but still predict correctness on multiple-choice Q&A. Preprint at https://arxiv.org/abs/2402.13213 (2024).
21. Yadkori, Y. A. et al. Mitigating LLM hallucinations via conformal abstention. Preprint at https://arxiv.org/abs/2405.01563 (2024).
22. Tomani, C., Chaudhuri, K., Evtimov, I., Cremers, D. & Ibrahim, M. Uncertainty-based abstention in LLMs improves safety and reduces hallucinations. Preprint at https://arxiv.org/abs/2404.10960 (2024).
23. Gold, J. I. & Shadlen, M. N. The neural basis of decision making. *Annu. Rev. Neurosci.* **30**, 535–574 (2007).
24. Yeung, N. & Summerfield, C. Metacognition in human decision-making: confidence and error monitoring. *Philos. Trans. R. Soc. B* **367**, 1310–1321 (2012).
25. Guo, C., Pleiss, G., Sun, Y. & Weinberger, K. Q. On calibration of modern neural networks. In *International Conference on Machine Learning* 1321–1330 (PMLR, 2017).
26. Lewis, P. et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. *Adv. Neural Inf. Process. Syst.* **33**, 9459–9474 (2020).
27. Bommasani, R. et al. On the opportunities and risks of foundation models. Preprint at https://arxiv.org/abs/2108.07258 (2021).
28. Reimers, N. & Gurevych, I. Sentence-BERT: sentence embeddings using Siamese BERT-networks. In *Proc. 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing* (eds Inui, K. et al.) 3982–3992 (Association for Computational Linguistics, 2019).
29. Turner, A. M. et al. Steering language models with activation engineering. Preprint at https://arxiv.org/abs/2308.10248 (2023).
30. Stolfo, A., Balachandran, V., Yousefi, S., Horvitz, E. & Nushi, B. Improving instruction-following in language models through activation steering. In *International Conference on Learning Representations* 55790–55823 (Association for Computational Linguistics, 2025).

31. VanderWeele, T. *Explanation in Causal Inference: Methods for Mediation and Interaction* (Oxford Univ. Press, 2015).
32. Wen, B., Howe, B. & Wang, L. L. Characterizing LLM abstention behavior in science QA with context perturbations. In *Findings of the Association for Computational Linguistics: EMNLP 2024* (eds Al-Onaizan, Y. et al.) 3437–3450 (Association for Computational Linguistics, 2024).
33. Sclar, M., Choi, Y., Tsvetkov, Y. & Suhr, A. Quantifying language models' sensitivity to spurious features in prompt design or: how I learned to start worrying about prompt formatting. In *International Conference on Learning Representations* 25055–25083 (Curran Associates, 2024).
34. Geng, J. et al. A survey of language model confidence estimation and calibration. Preprint at https://arxiv.org/abs/2311.08298 (2023).
35. Yoon, D. et al. Reasoning models better express their confidence. In *Advances in Neural Information Processing Systems* (eds Belgrave, D. et al.) 103869–103896 (Neural Information Processing Systems Foundation, 2026).
36. Steyvers, M., Belem, C. & Smyth, P. Improving metacognition and uncertainty communication in language models. Preprint at https://arxiv.org/abs/2510.05126 (2025).
37. Kumaran, D. et al. How do LLMs compute verbal confidence. In *Proc. 43rd International Conference on Machine Learning* (eds Emerson, T. et al.) (PMLR, 2026).
38. Niculescu-Mizil, A. & Caruana, R. Predicting good probabilities with supervised learning. In *Proc. 22nd International Conference on Machine Learning* 625–632 (ACM, 2005).
39. Kumaran, D. Reported confidence in LLMs tracks commitment more than correctness. Preprint at https://arxiv.org/abs/2606.29490 (2026).
40. Rimsky, N. et al. Steering llama 2 via contrastive activation addition. In *Proc. 62nd Annual Meeting of the Association for Computational Linguistics* 15504–15522 (Association for Computational Linguistics, 2024).
41. Kumaran, D., Patraucean, V., Osindero, S., Veličković, P. & Daw, N. How LLMs detect and correct their own errors: the role of internal confidence signals. Preprint at https://arxiv.org/abs/2604.22271 (2026).
42. Gandhi, K., Chakravarthy, A., Singh, A., Lile, N. & Goodman, N. D. Cognitive behaviors that enable self-improving reasoners, or, four habits of highly effective STaRs. Preprint at https://arxiv.org/abs/2503.01307 (2025).
43. Venhoff, C., Arcuschin, I., Torr, P., Conmy, A. & Nanda, N. Understanding reasoning in thinking language models via steering vectors. Preprint at https://arxiv.org/abs/2506.18167 (2025).
44. Tao, L. et al. Revisiting uncertainty estimation and calibration of large language models. Preprint at https://arxiv.org/abs/2505.23854 (2025).
45. Kadavath, S. et al. Language models (mostly) know what they know. Preprint at https://arxiv.org/abs/2207.05221 (2022).
46. Lindsey, J. Emergent introspective awareness in large language models (Anthropic, 2025); https://transformer-circuits.pub/2025/introspection/
47. Wei, J. et al. Measuring short-form factuality in large language models. Preprint at https://arxiv.org/abs/2411.04368 (2024).
48. Hua, T. T., Qin, A., Marks, S. & Nanda, N. Steering evaluation-aware language models to act like they are deployed. Preprint at https://arxiv.org/abs/2510.20487 (2025).
49. Kumaran, D. Code and data for Causal Abstention paper. *Open Science Framework* https://doi.org/10.17605/OSF.IO/RESW4 (2026).
50. Webb, T. W., Miyoshi, K., So, T. Y., Rajananda, S. & Lau, H. Natural statistics support a rational account of confidence biases. *Nat. Commun.* **14**, 3992 (2023).

Acknowledgements

We thank L. Smith for advice, C. Blundell for useful discussions and K. Stachenfeld for comments on an earlier version of the paper. We used Gemini to help improve the clarity and readability of part of the paper after the initial draft was completed. The revised text was rechecked by the authors for factuality.

Author contributions

D.K. conceived the project and experimental design. D.K. carried out the experiments and analysis with input from N.D. V.P., N.D., P.V. and S.O. advised on the project. D.K. wrote the paper, with input from N.D., V.P., P.V. and S.O.

Funding

Funding was provided by Google DeepMind.

Competing interests

The authors declare no competing interests

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s42256-026-01293-x.

Correspondence and requests for materials should be addressed to Dharshan Kumaran.

Peer review information *Nature Machine Intelligence* thanks the anonymous reviewer(s) for their contribution to the peer review of this work.

nature portfolio

Corresponding author(s): Dharshan Kumaran

Last updated by author(s): 3/7/2026

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	This is provided in the data availability statement
Data analysis	Analysis was performed using python in colab, using standard packages

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

\section*{Data Availability}
The dataset used in this study is available at \url{https://osf.io/resw4/overview?view_only=e96b5380edd04c64a37746b9b6ec7b49}.

\section*{Code Availability}

The full code required to carry out the experiments detailed in this manuscript is available at: \url{https://osf.io/resw4/overview?view_only=e96b5380edd04c64a37746b9b6ec7b49}.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	*Use the terms sex (biological attribute) and gender (shaped by social and cultural circumstances) carefully in order to avoid confusing both terms. Indicate if findings apply to only one sex or gender; describe whether sex and gender were considered in study design; whether sex and/or gender was determined based on self-reporting or assigned and methods used. Provide in the source data disaggregated sex and gender data, where this information has been collected, and if consent has been obtained for sharing of individual-level data; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex- and gender-based analyses where performed, justify reasons for lack of sex- and gender-based analysis.*
Reporting on race, ethnicity, or other socially relevant groupings	*Please specify the socially constructed or socially relevant categorization variable(s) used in your manuscript and explain why they were used. Please note that such variables should not be used as proxies for other socially constructed/relevant variables (for example, race or ethnicity should not be used as a proxy for socioeconomic status). Provide clear definitions of the relevant terms used, how they were provided (by the participants/respondents, the researchers, or third parties), and the method(s) used to classify people into the different categories (e.g. self-report, census or administrative data, social media data, etc.) Please provide details about how you controlled for confounding variables in your analyses.*
Population characteristics	*Describe the covariate-relevant population characteristics of the human research participants (e.g. age, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."*
Recruitment	*Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.*
Ethics oversight	*Identify the organization(s) that approved the study protocol.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	*Describe how sample size was determined, detailing any statistical methods used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.*
Data exclusions	*Describe any data exclusions. If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Replication	*Describe the measures taken to verify the reproducibility of the experimental findings. If all attempts at replication were successful, confirm this OR if there are any findings that were not replicated or cannot be reproduced, note this and describe why.*
Randomization	*Describe how samples/organisms/participants were allocated into experimental groups. If allocation was not random, describe how covariates were controlled OR if this is not relevant to your study, explain why.*
Blinding	*Describe whether the investigators were blinded to group allocation during data collection and/or analysis. If blinding was not possible, describe why OR explain why blinding was not relevant to your study.*

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	*Briefly describe the study type including whether data are quantitative, qualitative, or mixed-methods (e.g. qualitative cross-sectional, quantitative experimental, mixed-methods case study).*
Research sample	*State the research sample (e.g. Harvard university undergraduates, villagers in rural India) and provide relevant demographic*

Research sample	*information (e.g. age, sex) and indicate whether the sample is representative. Provide a rationale for the study sample chosen. For studies involving existing datasets, please describe the dataset and source.*
Sampling strategy	*Describe the sampling procedure (e.g. random, snowball, stratified, convenience). Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient. For qualitative data, please indicate whether data saturation was considered, and what criteria were used to decide that no further sampling was needed.*
Data collection	*Provide details about the data collection procedure, including the instruments or devices used to record the data (e.g. pen and paper, computer, eye tracker, video or audio equipment) whether anyone was present besides the participant(s) and the researcher, and whether the researcher was blind to experimental condition and/or the study hypothesis during data collection.*
Timing	*Indicate the start and stop dates of data collection. If there is a gap between collection periods, state the dates for each sample cohort.*
Data exclusions	*If no data were excluded from the analyses, state so OR if data were excluded, provide the exact number of exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Non-participation	*State how many participants dropped out/declined participation and the reason(s) given OR provide response rate OR state that no participants dropped out/declined participation.*
Randomization	*If participants were not allocated into experimental groups, state so OR describe how participants were allocated to groups, and if allocation was not random, describe how covariates were controlled.*

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	*Briefly describe the study. For quantitative data include treatment factors and interactions, design structure (e.g. factorial, nested, hierarchical), nature and number of experimental units and replicates.*
Research sample	*Describe the research sample (e.g. a group of tagged Passer domesticus, all Stenocereus thurberi within Organ Pipe Cactus National Monument), and provide a rationale for the sample choice. When relevant, describe the organism taxa, source, sex, age range and any manipulations. State what population the sample is meant to represent when applicable. For studies involving existing datasets, describe the data and its source.*
Sampling strategy	*Note the sampling procedure. Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.*
Data collection	*Describe the data collection procedure, including who recorded the data and how.*
Timing and spatial scale	*Indicate the start and stop dates of data collection, noting the frequency and periodicity of sampling and providing a rationale for these choices. If there is a gap between collection periods, state the dates for each sample cohort. Specify the spatial scale from which the data are taken*
Data exclusions	*If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Reproducibility	*Describe the measures taken to verify the reproducibility of experimental findings. For each experiment, note whether any attempts to repeat the experiment failed OR state that all attempts to repeat the experiment were successful.*
Randomization	*Describe how samples/organisms/participants were allocated into groups. If allocation was not random, describe how covariates were controlled. If this is not relevant to your study, explain why.*
Blinding	*Describe the extent of blinding used during data acquisition and analysis. If blinding was not possible, describe why OR explain why blinding was not relevant to your study.*

Did the study involve field work? ☐ Yes ☐ No

Field work, collection and transport

Field conditions	*Describe the study conditions for field work, providing relevant parameters (e.g. temperature, rainfall).*
Location	*State the location of the sampling or experiment, providing relevant parameters (e.g. latitude and longitude, elevation, water depth).*
Access & import/export	*Describe the efforts you have made to access habitats and to collect and import/export your samples in a responsible manner and in compliance with local, national and international laws, noting any permits that were obtained (give the name of the issuing authority, the date of issue, and any identifying information).*

Disturbance | Describe any disturbance caused by the study and how it was minimized.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☐ Antibodies
☐	☐ Eukaryotic cell lines
☐	☐ Palaeontology and archaeology
☐	☐ Animals and other organisms
☐	☐ Clinical data
☐	☐ Dual use research of concern
☐	☐ Plants

Methods

n/a	Involved in the study
☐	☐ ChIP-seq
☐	☐ Flow cytometry
☐	☐ MRI-based neuroimaging

Antibodies

Antibodies used | *Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.*

Validation | *Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.*

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s) | *State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.*

Authentication | *Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.*

Mycoplasma contamination | *Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.*

Commonly misidentified lines (See ICLAC register) | *Name any commonly misidentified cell lines used in the study and provide a rationale for their use.*

Palaeontology and Archaeology

Specimen provenance | *Provide provenance information for specimens and describe permits that were obtained for the work (including the name of the issuing authority, the date of issue, and any identifying information). Permits should encompass collection and, where applicable, export.*

Specimen deposition | *Indicate where the specimens have been deposited to permit free access by other researchers.*

Dating methods | *If new dates are provided, describe how they were obtained (e.g. collection, storage, sample pretreatment and measurement), where they were obtained (i.e. lab name), the calibration program and the protocol for quality assurance OR state that no new dates are provided.*

☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.

Ethics oversight | *Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	*For laboratory animals, report species, strain and age OR state that the study did not involve laboratory animals.*
Wild animals	*Provide details on animals observed in or captured in the field; report species and age where possible. Describe how animals were caught and transported and what happened to captive animals after the study (if killed, explain why and describe method; if released, say where and when) OR state that the study did not involve wild animals.*
Reporting on sex	*Indicate if findings apply to only one sex; describe whether sex was considered in study design, methods used for assigning sex. Provide data disaggregated for sex where this information has been collected in the source data as appropriate; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex-based analyses where performed, justify reasons for lack of sex-based analysis.*
Field-collected samples	*For laboratory work with field-collected samples, describe all relevant parameters such as housing, maintenance, temperature, photoperiod and end-of-experiment protocol OR state that the study did not involve samples collected from the field.*
Ethics oversight	*Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	*Provide the trial registration number from ClinicalTrials.gov or an equivalent agency.*
Study protocol	*Note where the full trial protocol can be accessed OR if not available, explain why.*
Data collection	*Describe the settings and locales of data collection, noting the time periods of recruitment and data collection.*
Outcomes	*Describe how you pre-defined primary and secondary outcome measures and how you assessed these measures.*

Dual use research of concern

Policy information about dual use research of concern

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes	
☐	☐	Public health
☐	☐	National security
☐	☐	Crops and/or livestock
☐	☐	Ecosystems
☐	☐	Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes	
☐	☐	Demonstrate how to render a vaccine ineffective
☐	☐	Confer resistance to therapeutically useful antibiotics or antiviral agents
☐	☐	Enhance the virulence of a pathogen or render a nonpathogen virulent
☐	☐	Increase transmissibility of a pathogen
☐	☐	Alter the host range of a pathogen
☐	☐	Enable evasion of diagnostic/detection modalities
☐	☐	Enable the weaponization of a biological agent or toxin
☐	☐	Any other potentially harmful combination of experiments and agents

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*

ChIP-seq

Data deposition

☐ Confirm that both raw and final processed data have been deposited in a public database such as GEO.

☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links *May remain private before publication.*	*For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.*
Files in database submission	*Provide a list of all files available in the database submission.*
Genome browser session (e.g. UCSC)	*Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.*

Methodology

Replicates	*Describe the experimental replicates, specifying number, type and replicate agreement.*
Sequencing depth	*Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.*
Antibodies	*Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.*
Peak calling parameters	*Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.*
Data quality	*Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.*
Software	*Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.*

Flow Cytometry

Plots

Confirm that:

- [ ] The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- [ ] The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- [ ] All plots are contour plots with outliers or pseudocolor plots.
- [ ] A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	*Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.*
Instrument	*Identify the instrument used for data collection, specifying make and model number.*
Software	*Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.*
Cell population abundance	*Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.*
Gating strategy	*Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.*

- [ ] Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type	*Indicate task or resting state; event-related or block design.*
Design specifications	*Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.*
Behavioral performance measures	*State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).*

Acquisition

Imaging type(s)	*Specify: functional, structural, diffusion, perfusion.*
Field strength	*Specify in Tesla*
Sequence & imaging parameters	*Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.*
Area of acquisition	*State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.*

Diffusion MRI ☐ Used ☐ Not used

Preprocessing

Preprocessing software	*Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).*
Normalization	*If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.*
Normalization template	*Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.*
Noise and artifact removal	*Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).*

Volume censoring | *Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.*

Statistical modeling & inference

Model type and settings | *Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).*

Effect(s) tested | *Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.*

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference | *Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.*

(See Eklund et al. 2016)

Correction | *Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).*

Models & analysis

n/a	Involved in the study
☐	☐ Functional and/or effective connectivity
☐	☐ Graph analysis
☐	☐ Multivariate modeling or predictive analysis

Functional and/or effective connectivity | *Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).*

Graph analysis | *Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).*

Multivariate modeling and predictive analysis | *Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.*